

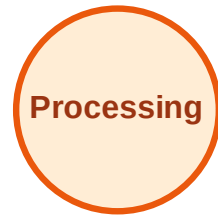
# BFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



Complete



Processing



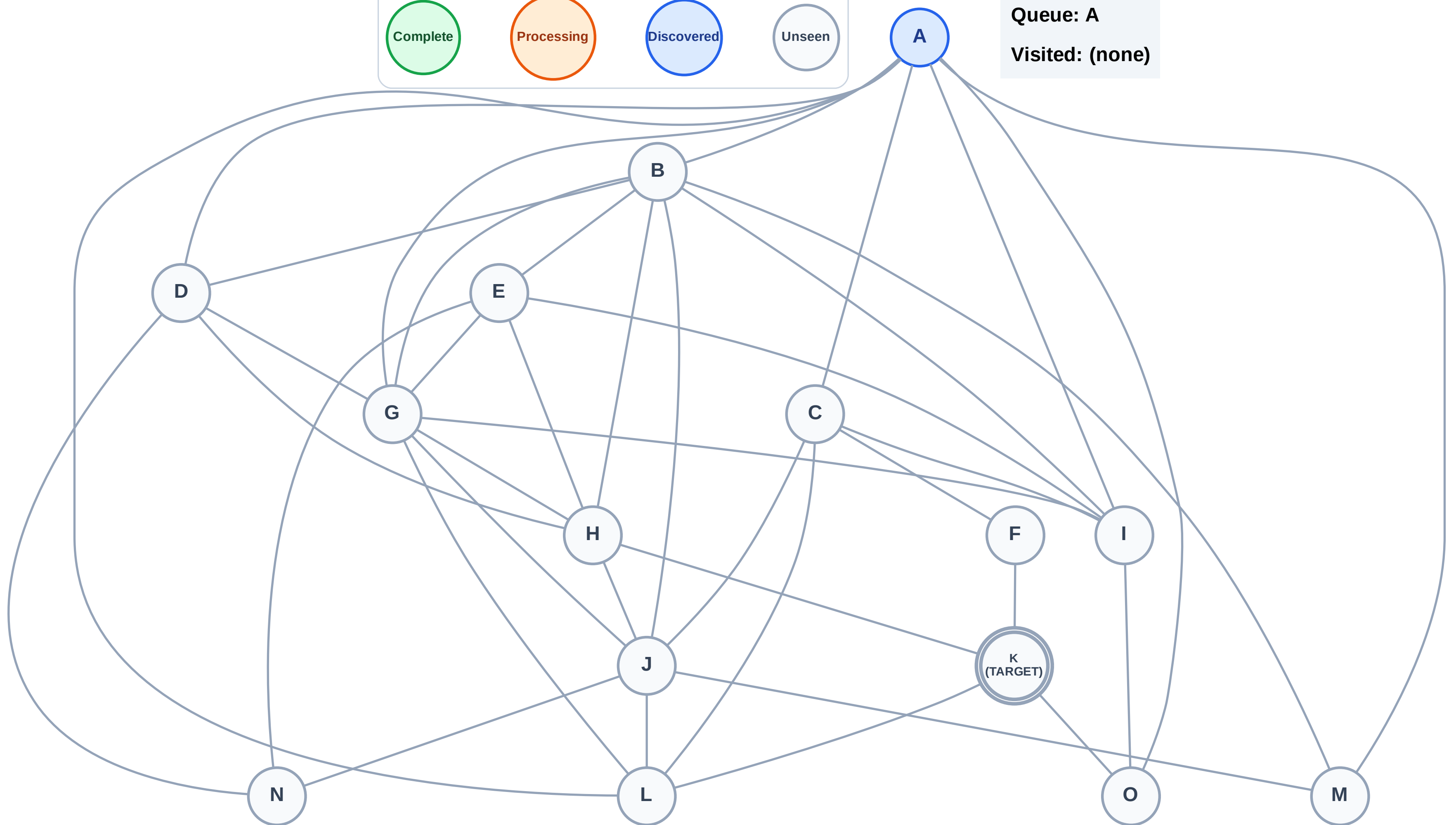
Discovered



Unseen

Queue: A

Visited: (none)



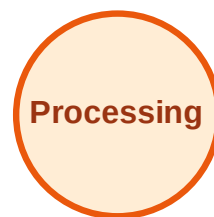
# BFS Traversal

Step 2: Process node A

## Legend



Complete



Processing



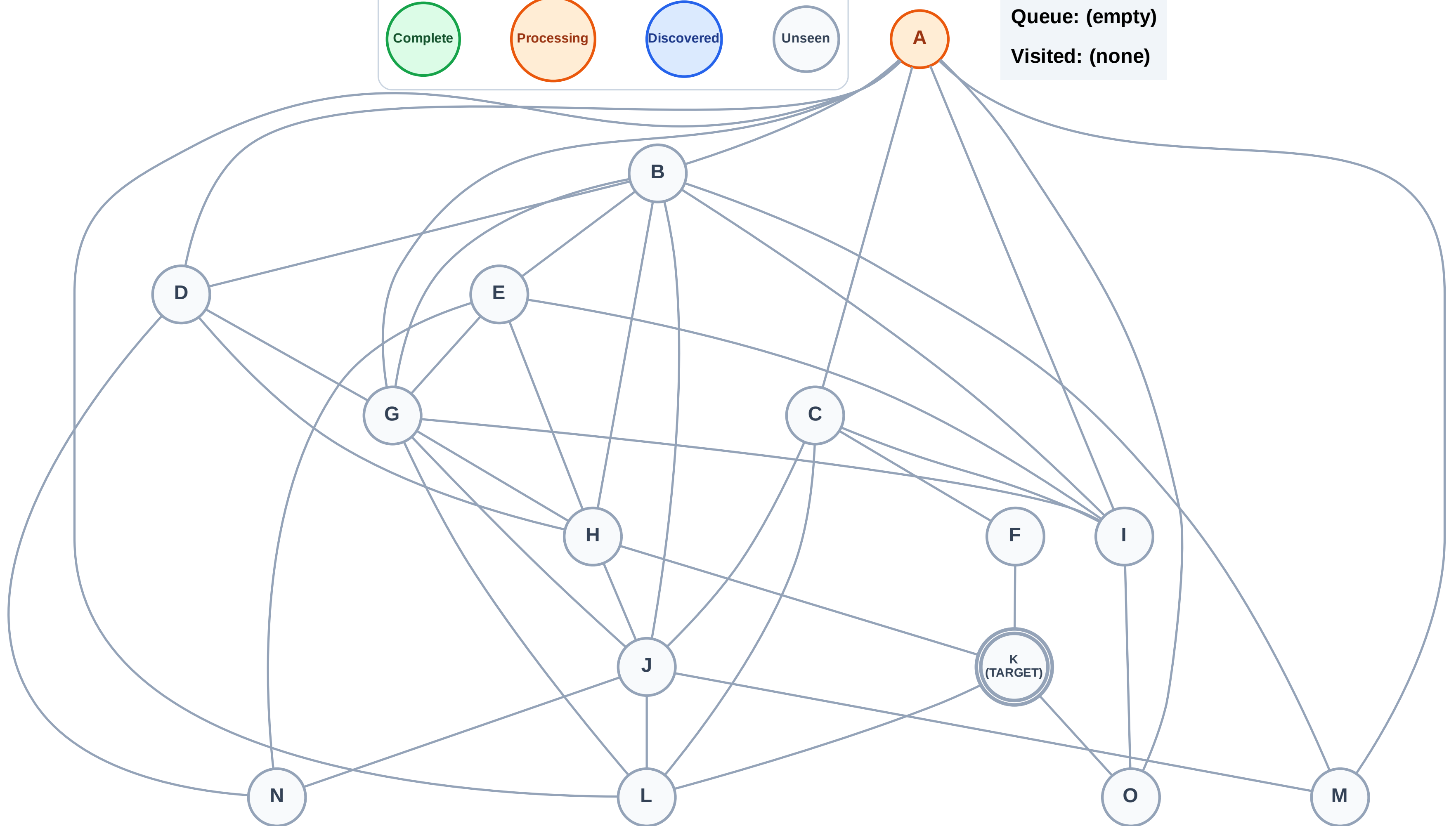
Discovered



Unseen

Queue: (empty)

Visited: (none)



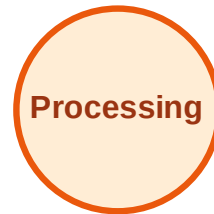
# BFS Traversal

Step 3: Discover B, C, D, G, I, L, M, O from A

## Legend



Complete



Processing



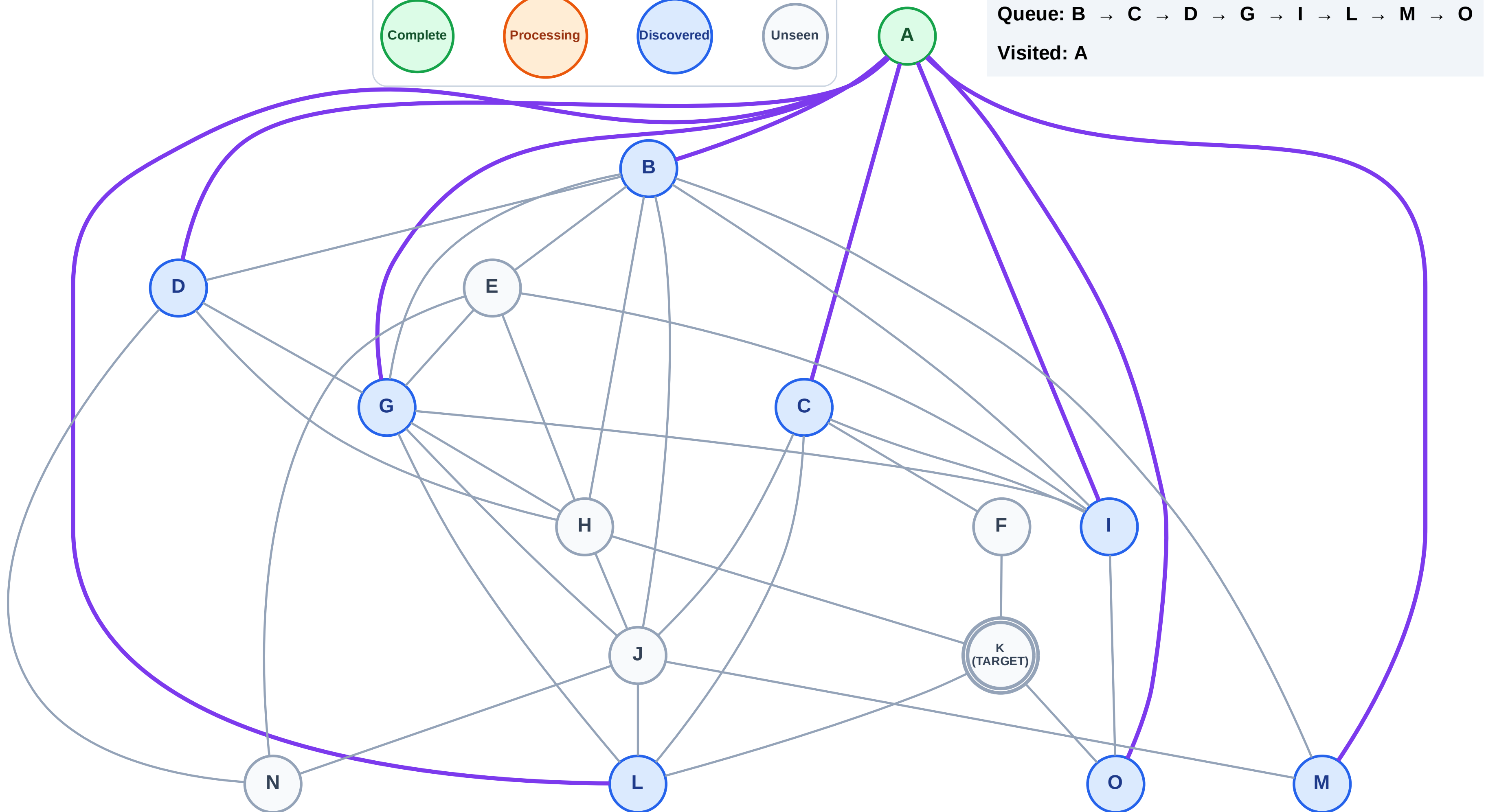
Discovered



Unseen

Queue: B → C → D → G → I → L → M → O

Visited: A



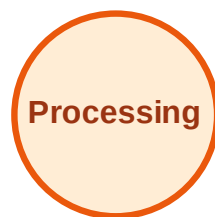
# BFS Traversal

Step 4: Process node B

## Legend



Complete



Processing



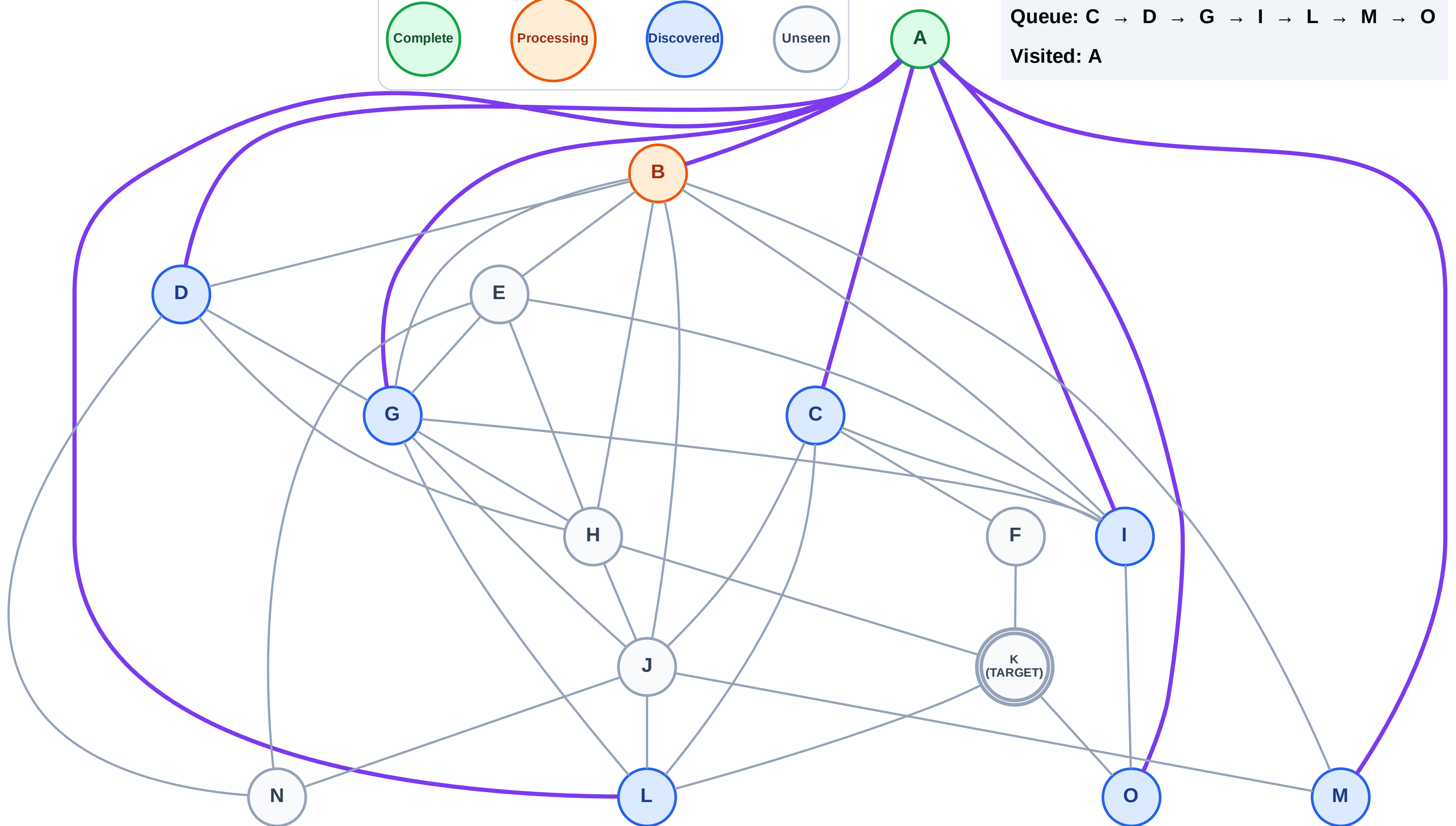
Discovered



Unseen

Queue: C → D → G → I → L → M → O

Visited: A



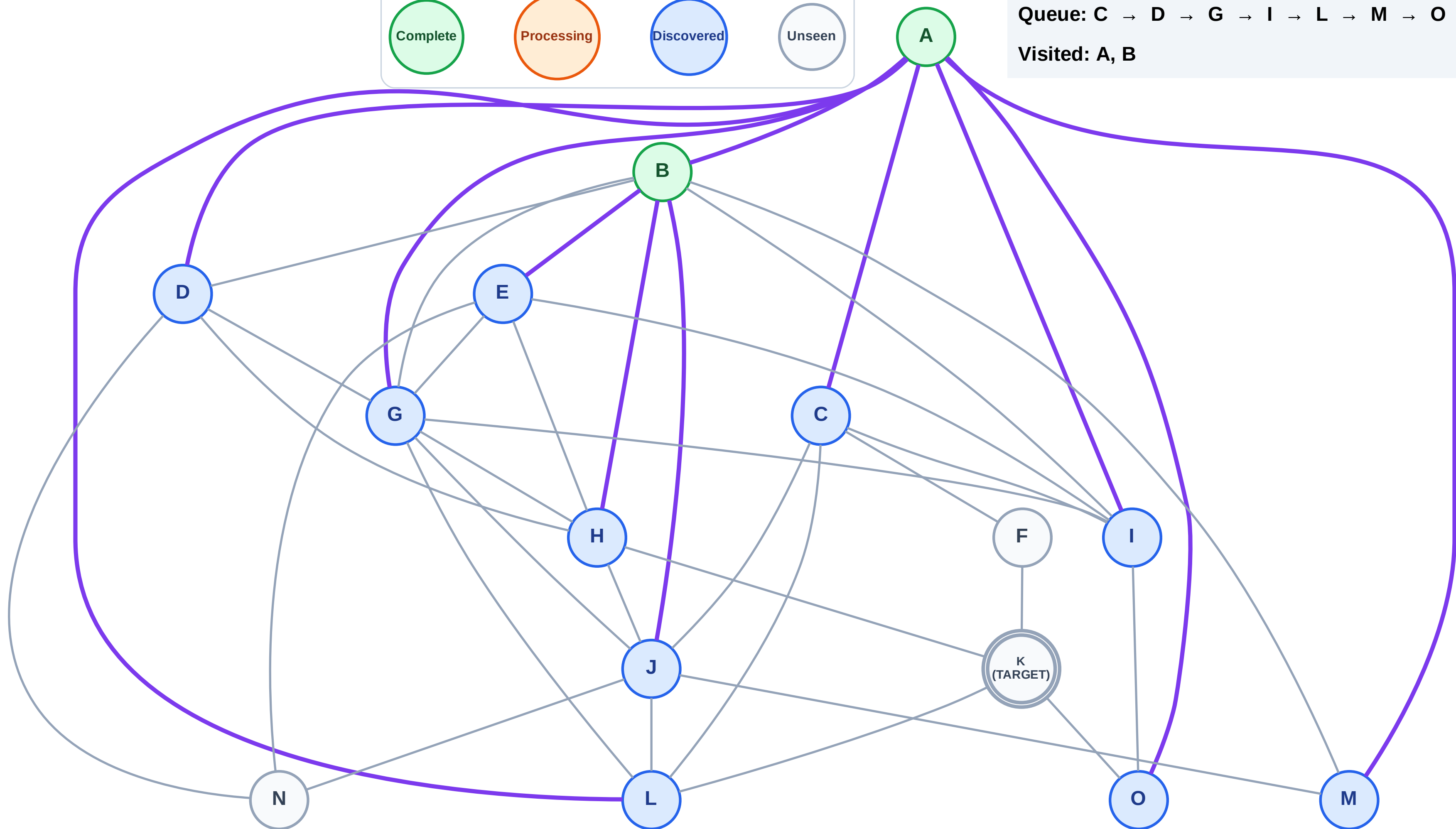
# BFS Traversal

Step 5: Discover E, H, J from B

## Legend



Queue: C → D → G → I → L → M → O → E → H → J  
Visited: A, B



# BFS Traversal

Step 6: Process node C

Legend

Complete

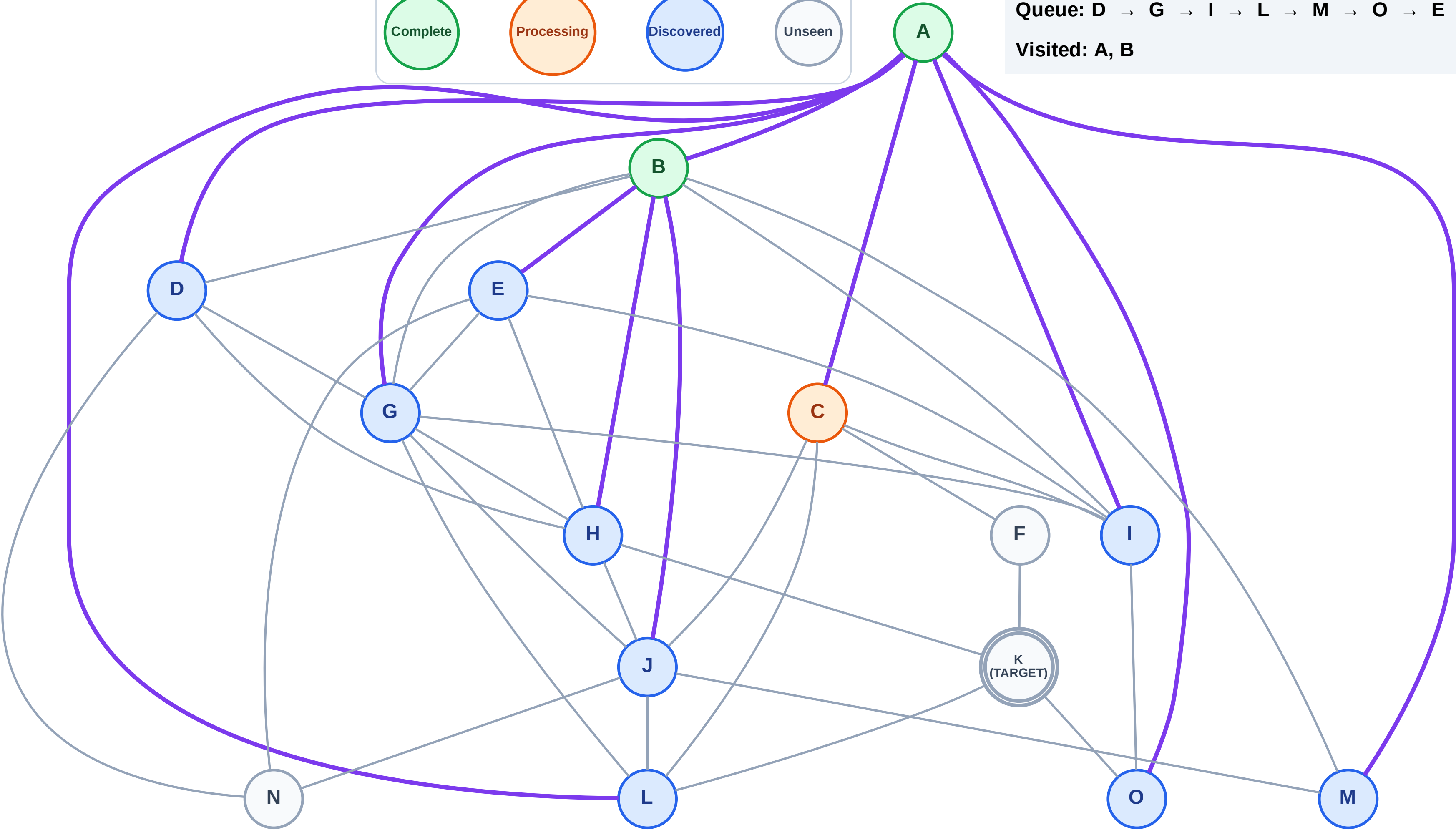
Processing

Discovered

Unseen

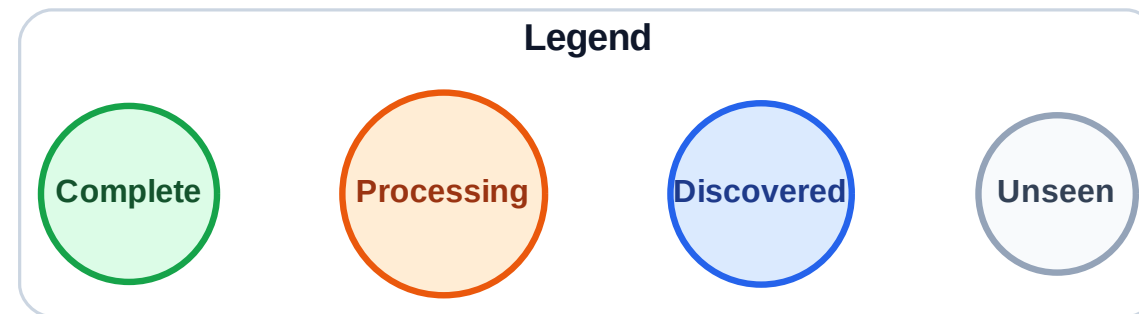
Queue: D → G → I → L → M → O → E → H → J

Visited: A, B



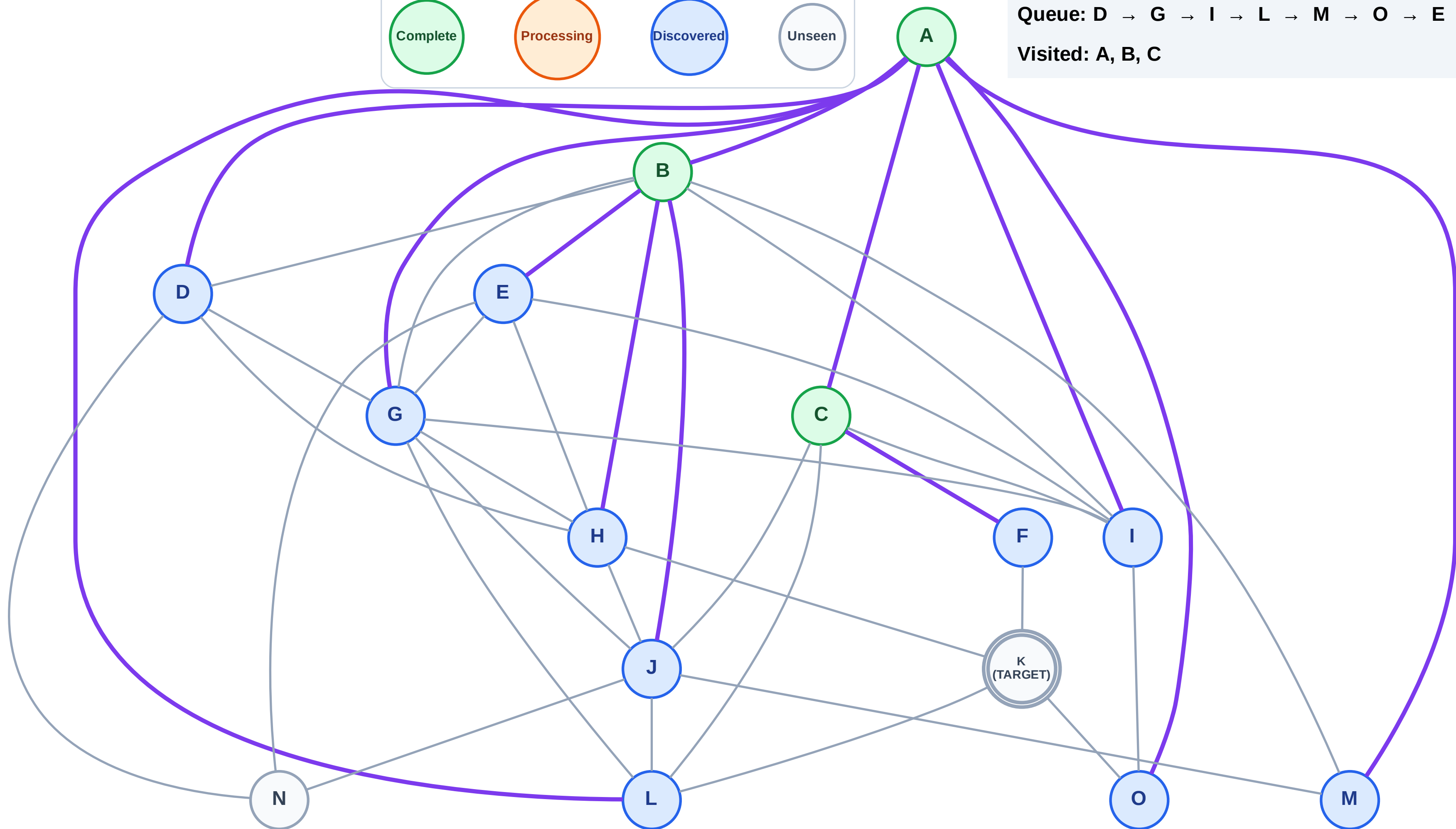
# BFS Traversal

Step 7: Discover F from C



Queue: D → G → I → L → M → O → E → H → J → F

Visited: A, B, C



# BFS Traversal

Step 8: Process node D

Legend

Complete

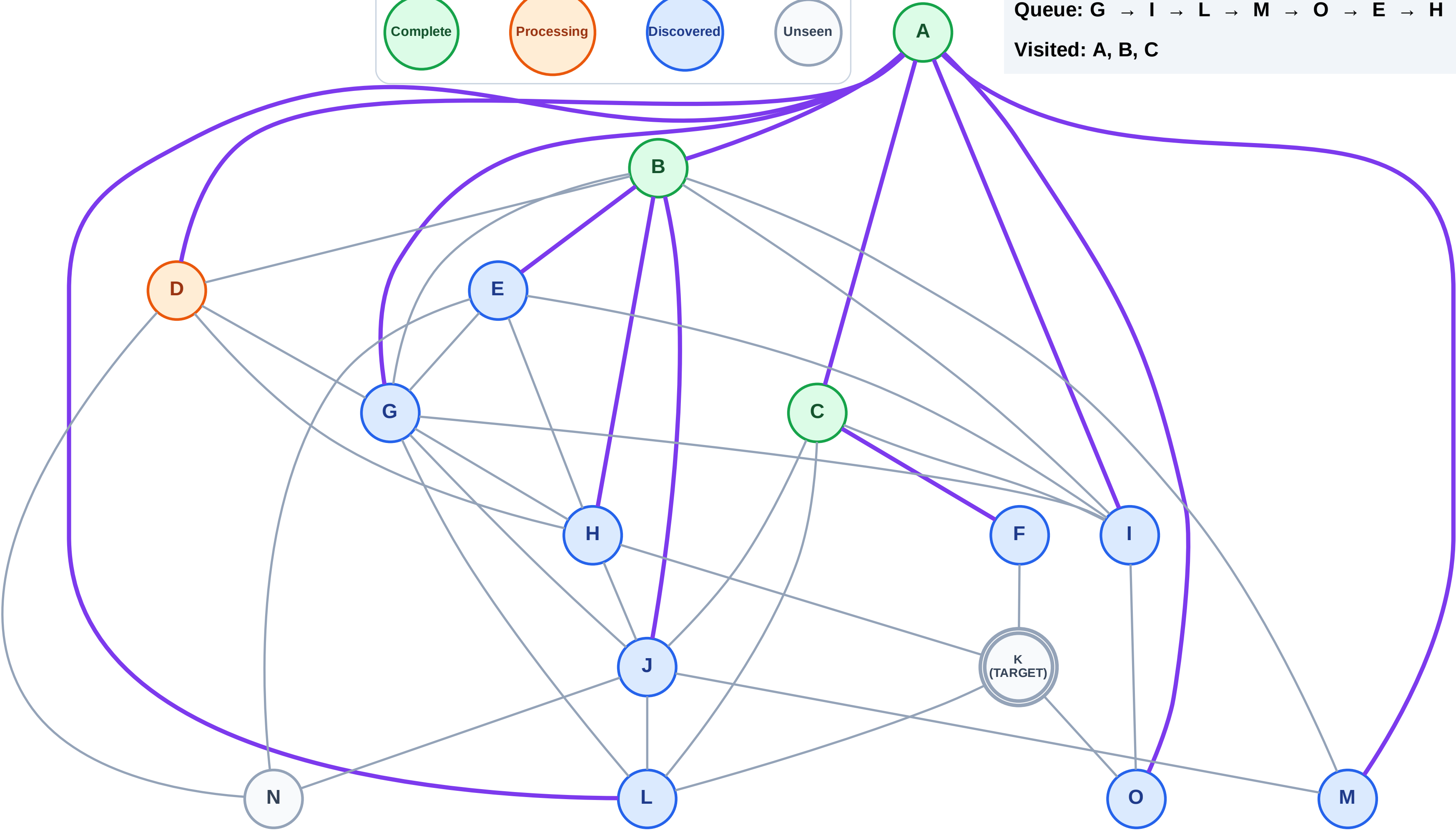
Processing

Discovered

Unseen

Queue: G → I → L → M → O → E → H → J → F

Visited: A, B, C



# BFS Traversal

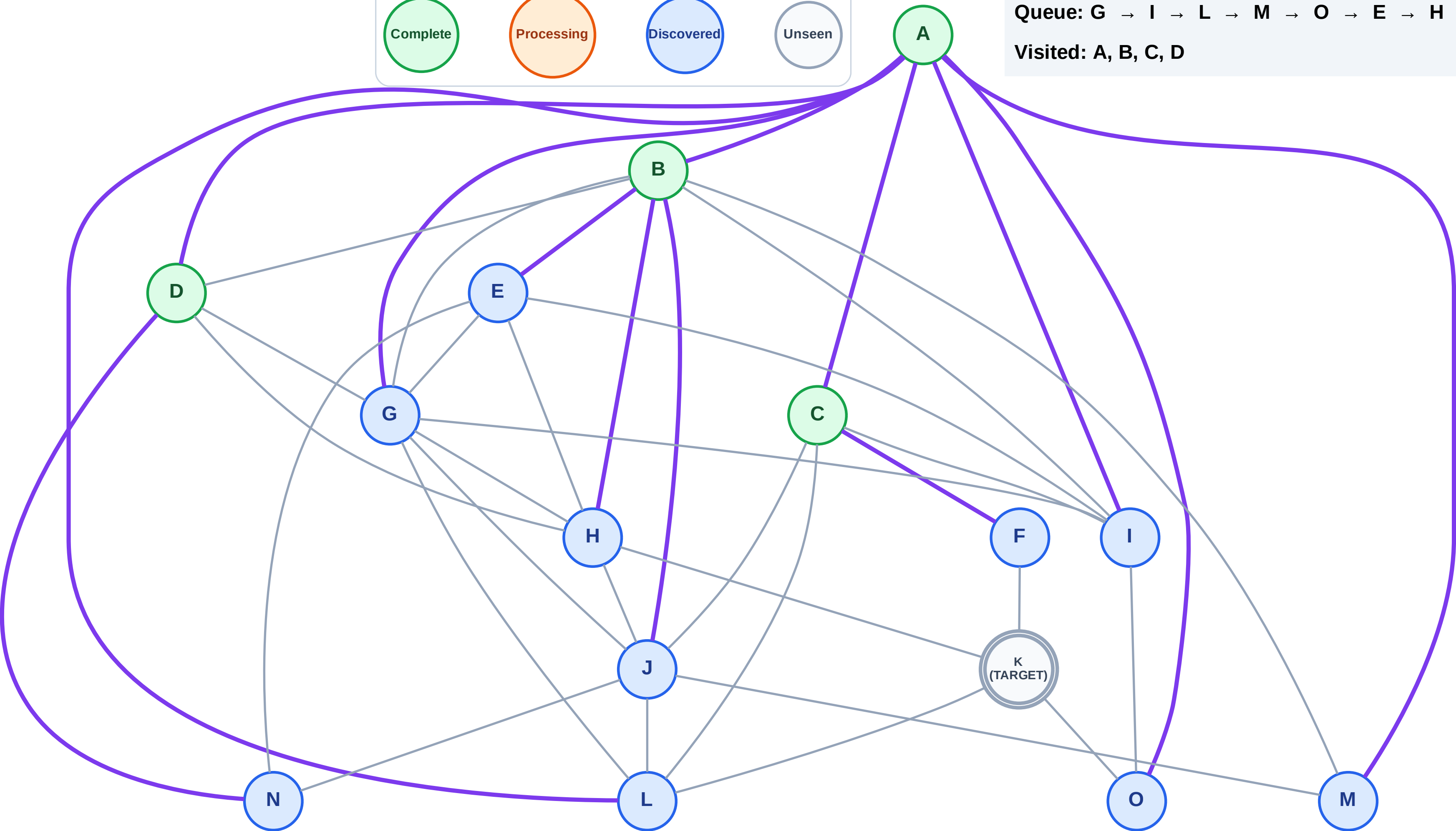
Step 9: Discover N from D

## Legend



Queue: G → I → L → M → O → E → H → J → F → N

Visited: A, B, C, D



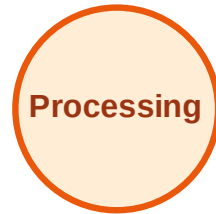
# BFS Traversal

Step 10: Process node G

## Legend



Complete



Processing



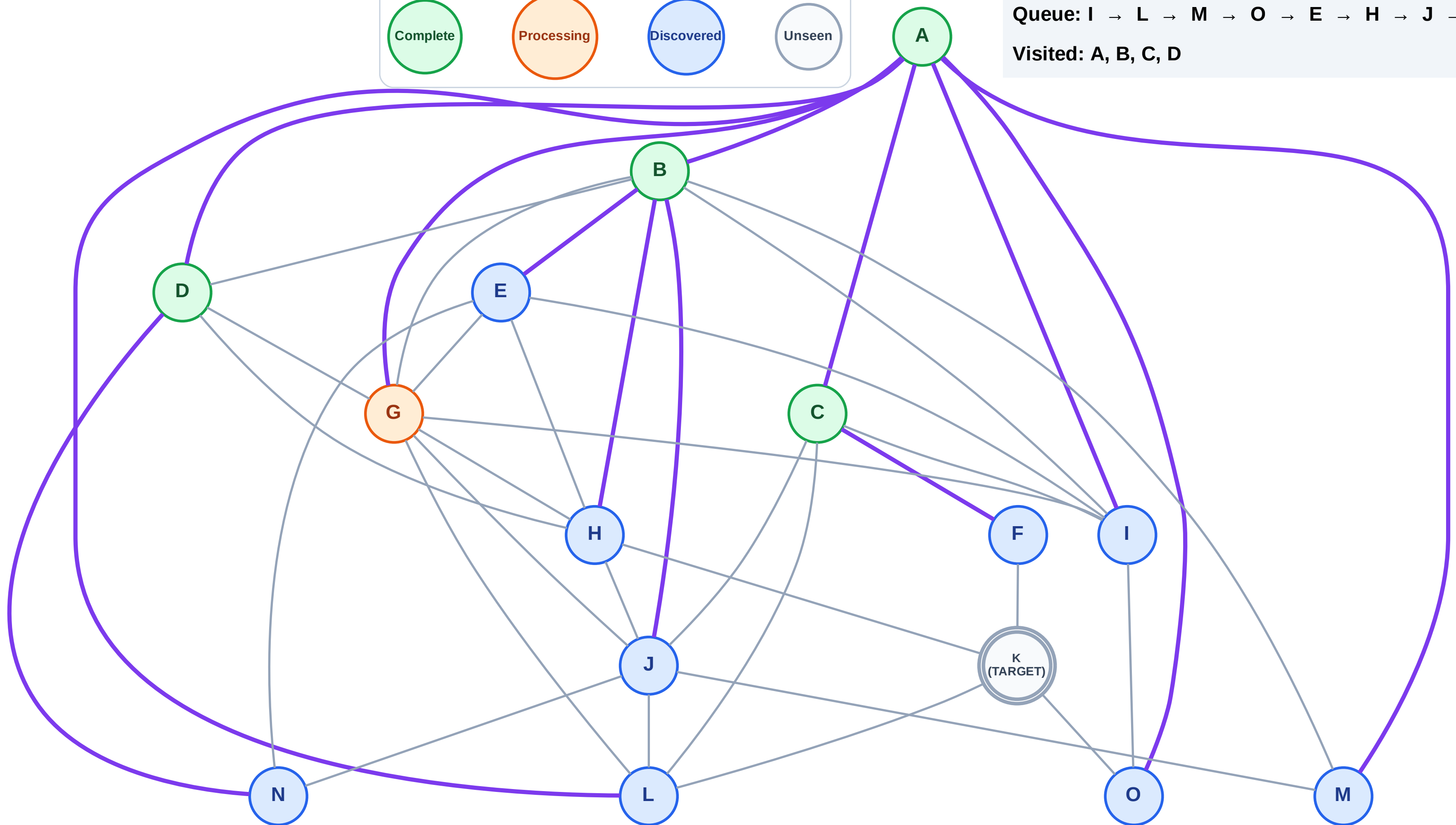
Discovered



Unseen

Queue: I → L → M → O → E → H → J → F → N

Visited: A, B, C, D



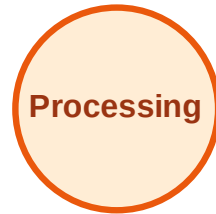
# BFS Traversal

Step 11: No new neighbors from G

## Legend



Complete



Processing



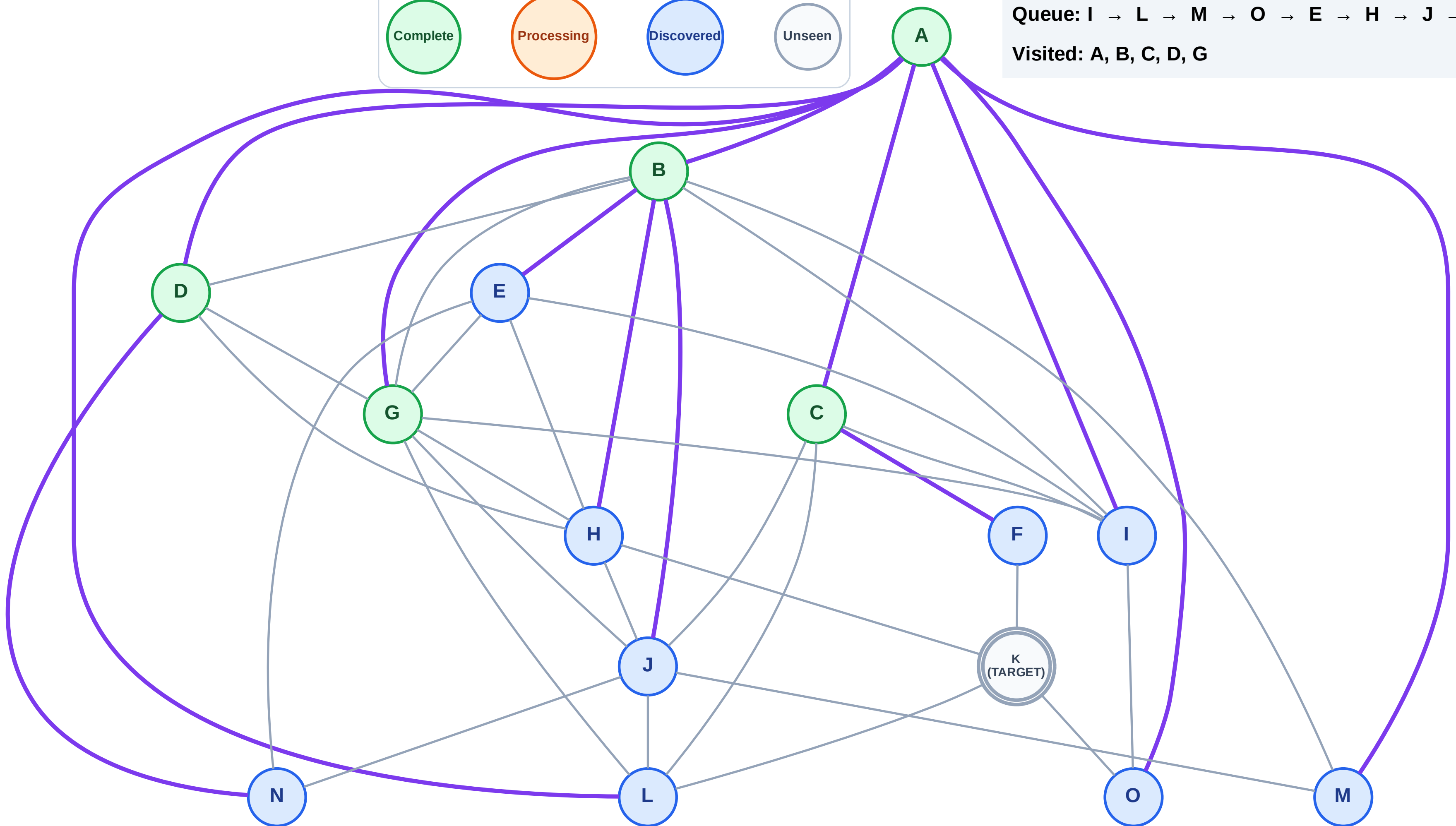
Discovered



Unseen

Queue: I → L → M → O → E → H → J → F → N

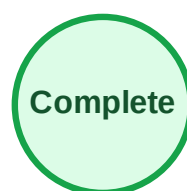
Visited: A, B, C, D, G



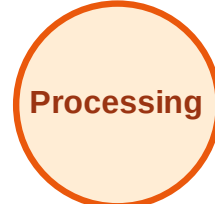
# BFS Traversal

Step 12: Process node I

## Legend



Complete



Processing



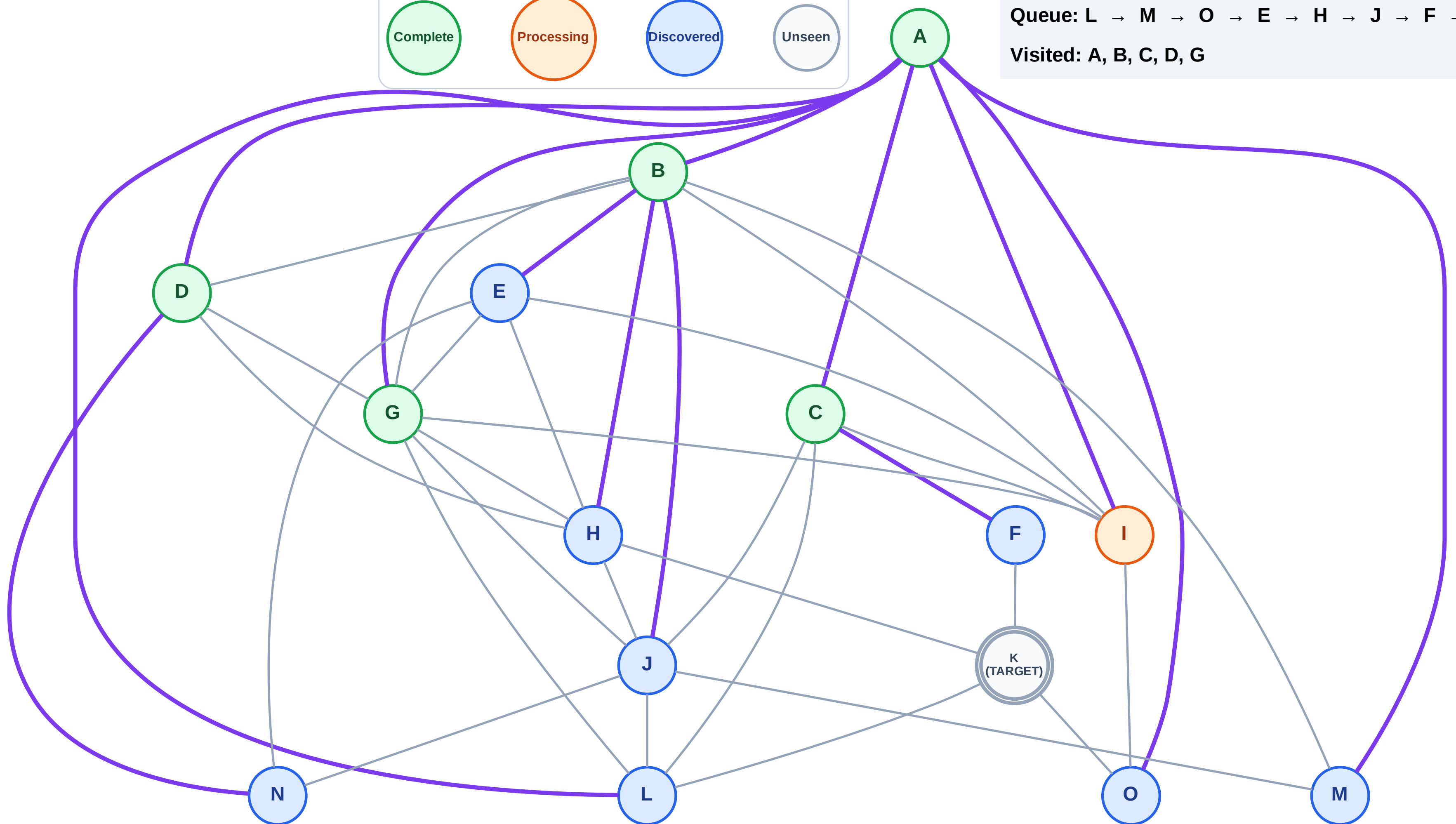
Discovered



Unseen

Queue: L → M → O → E → H → J → F → N

Visited: A, B, C, D, G



# BFS Traversal

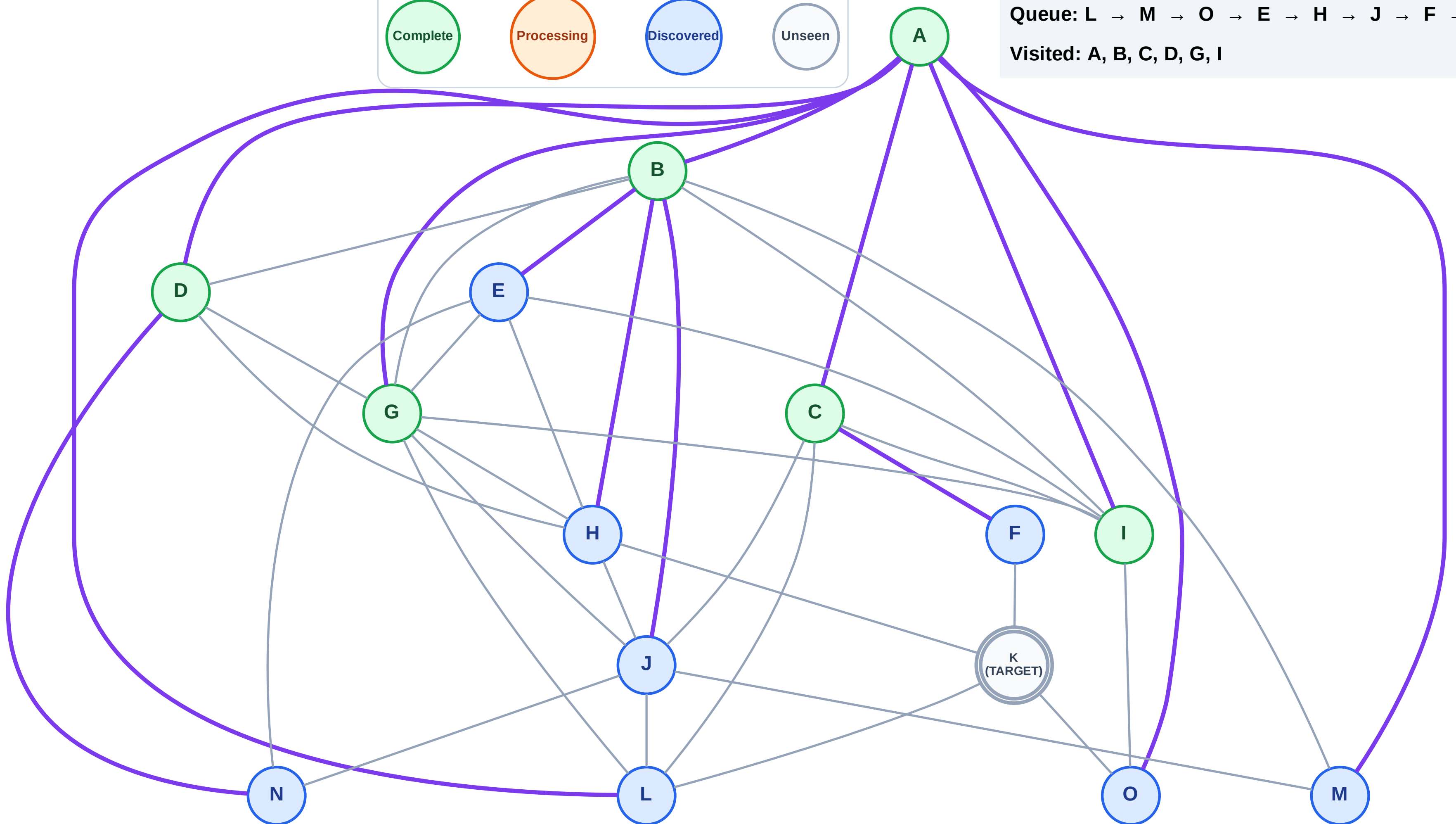
Step 13: No new neighbors from I

## Legend



Queue: L → M → O → E → H → J → F → N

Visited: A, B, C, D, G, I



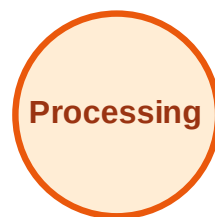
# BFS Traversal

Step 14: Process node L

## Legend



Complete



Processing



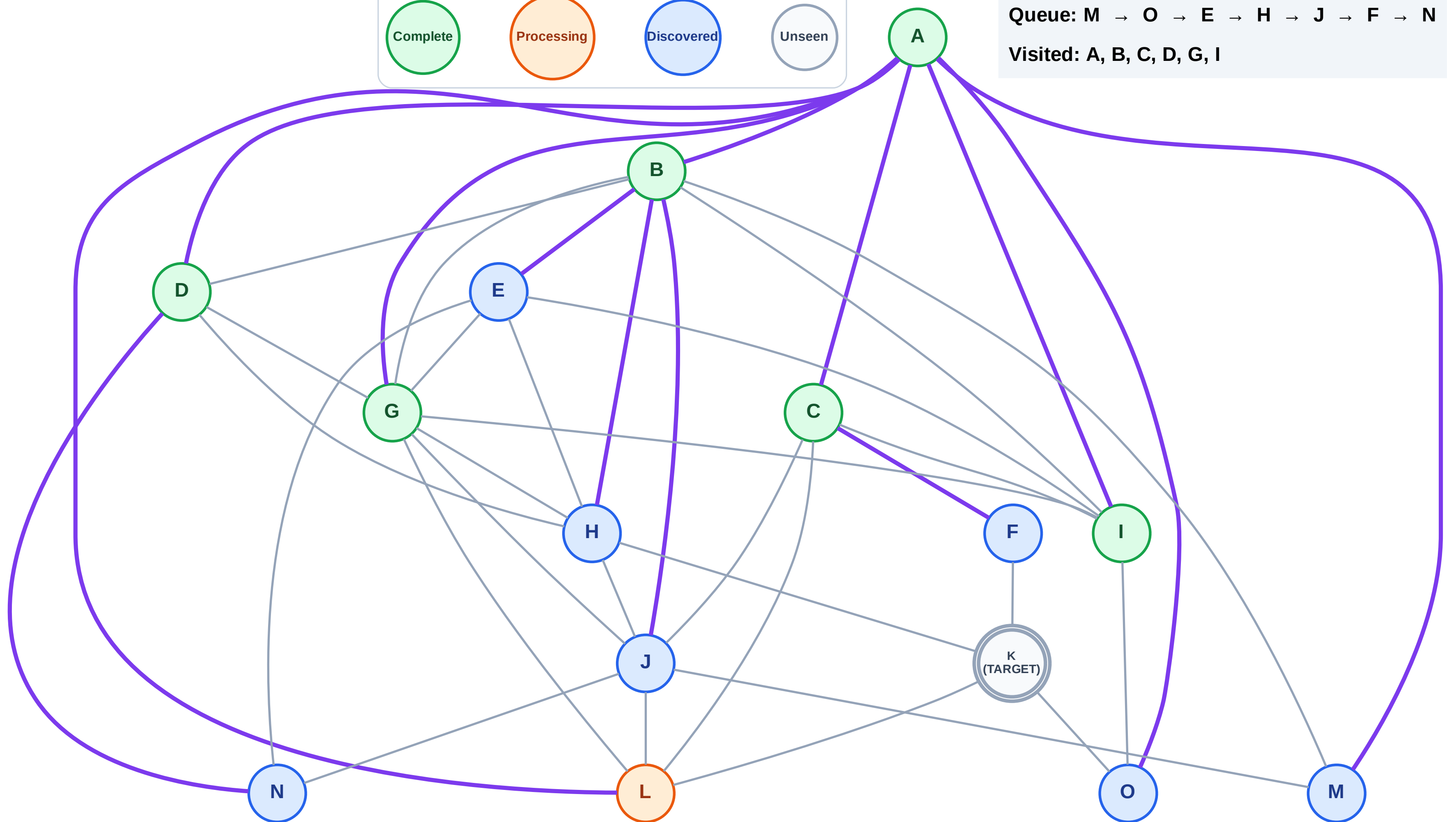
Discovered



Unseen

Queue: M → O → E → H → J → F → N

Visited: A, B, C, D, G, I



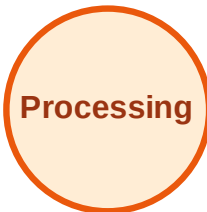
# BFS Traversal

Step 15: Discover K from L

## Legend



Complete



Processing



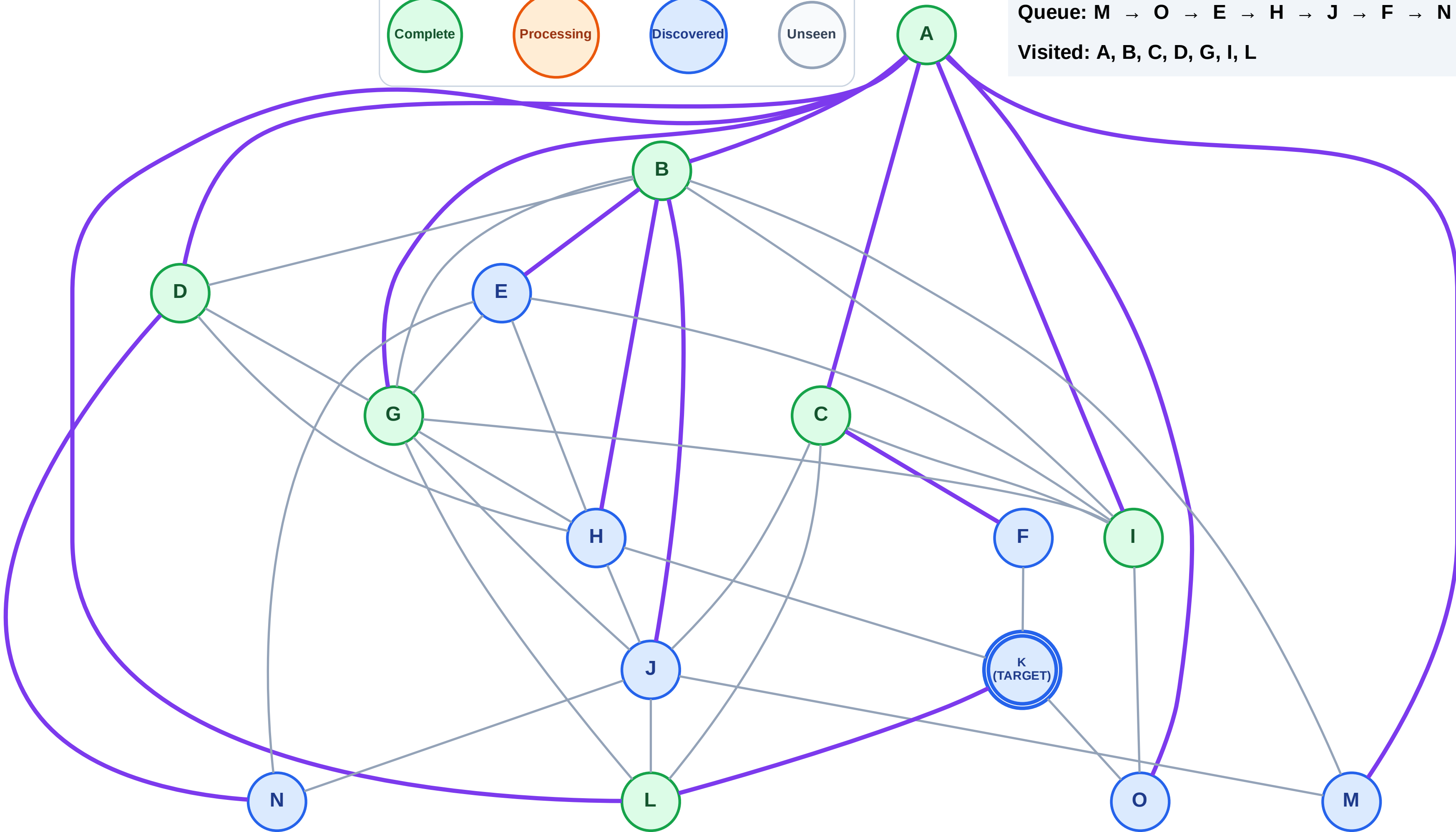
Discovered



Unseen

Queue: M → O → E → H → J → F → N → K

Visited: A, B, C, D, G, I, L



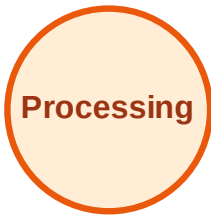
# BFS Traversal

Step 16: Process node M

## Legend



Complete



Processing



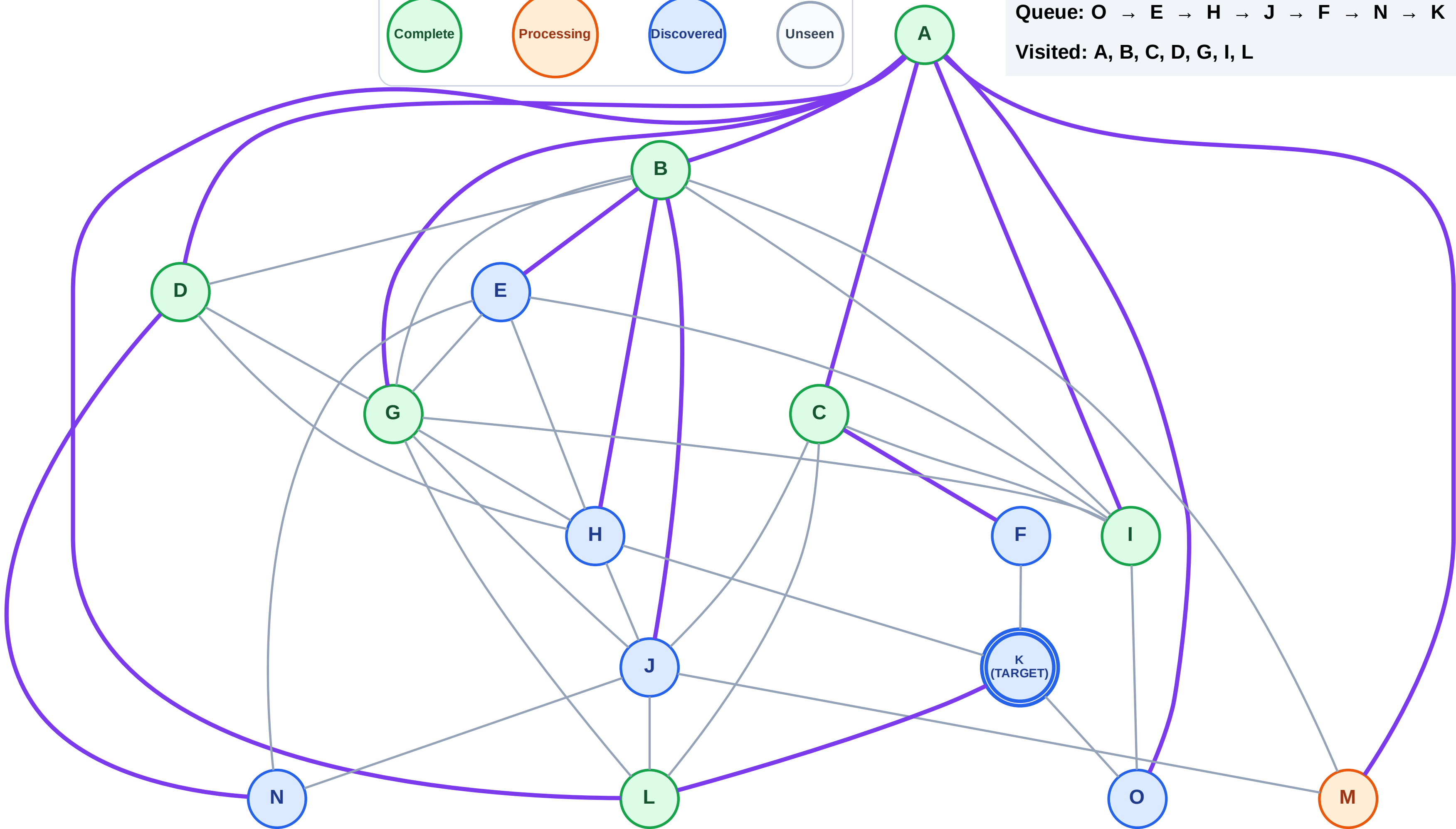
Discovered



Unseen

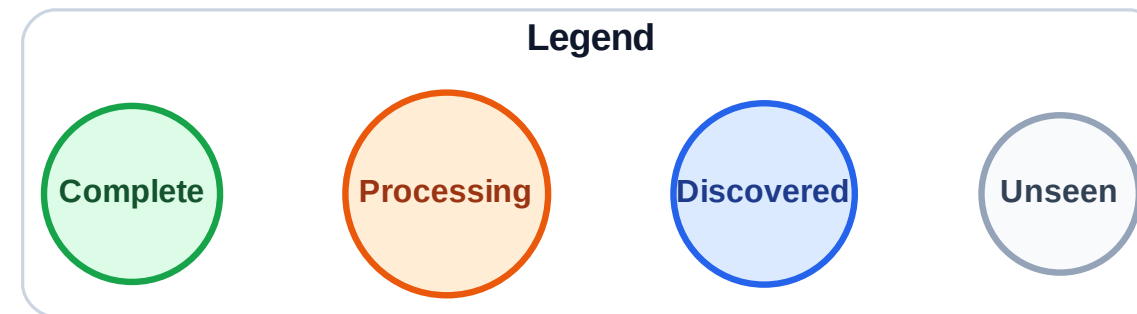
Queue: O → E → H → J → F → N → K

Visited: A, B, C, D, G, I, L



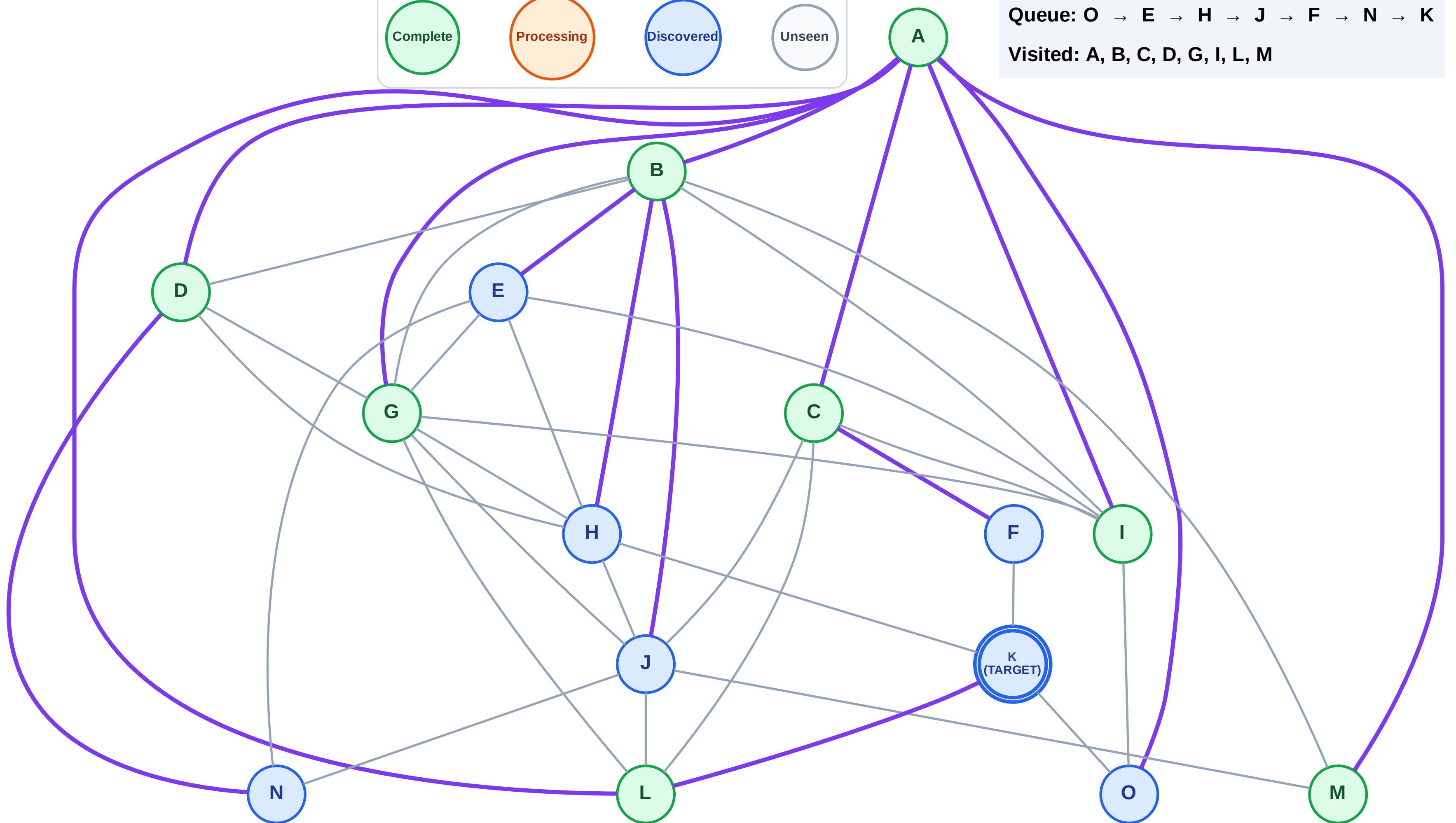
Step 17: No new neighbors from M

Step 17: No new neighbors from M



**Queue: O → E → H → J → F → N → K**

**Visited: A, B, C, D, G, I, L, M**



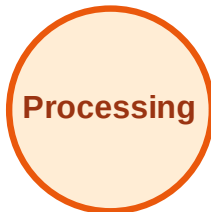
# BFS Traversal

Step 18: Process node O

## Legend



Complete



Processing



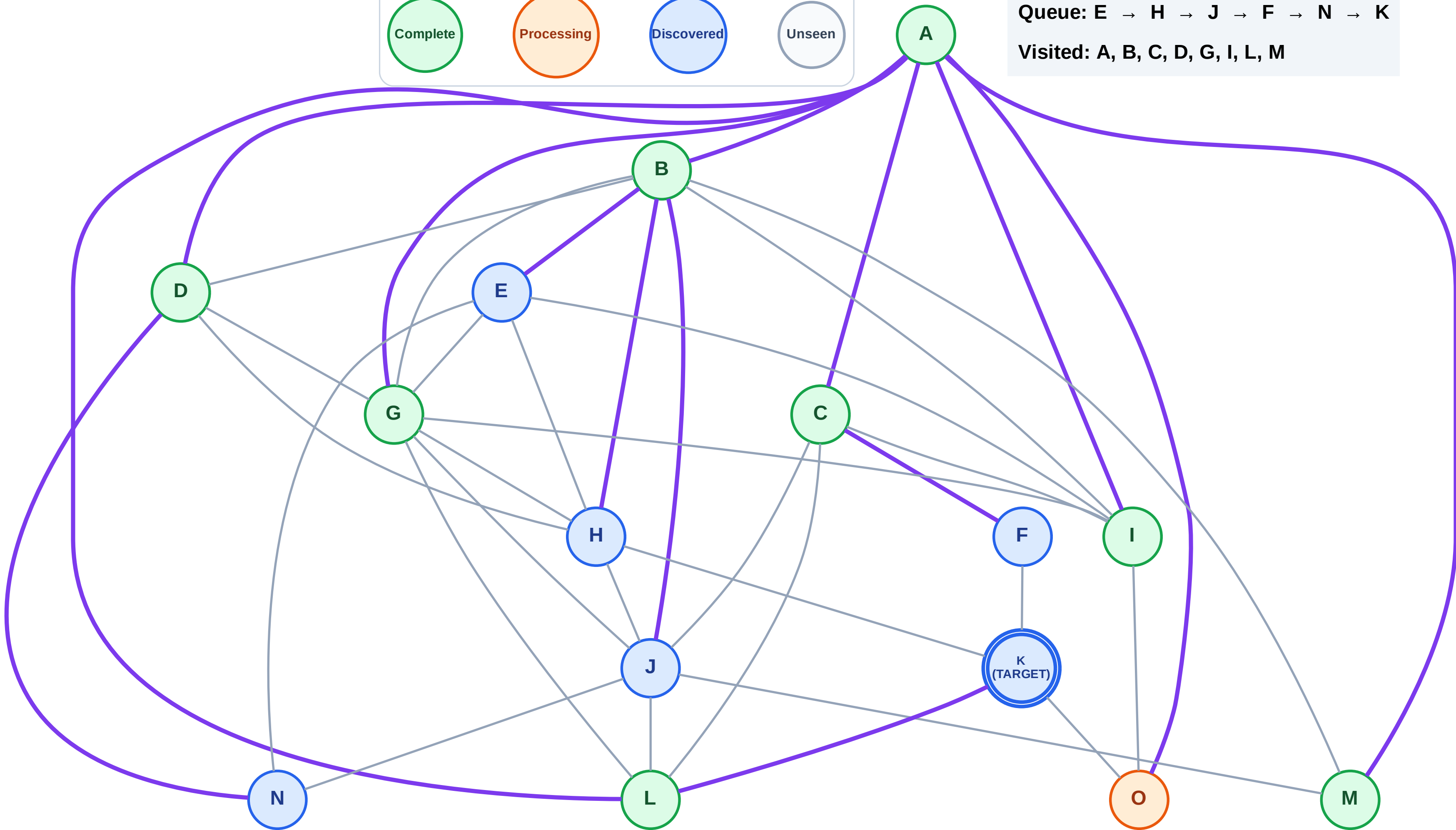
Discovered



Unseen

Queue: E → H → J → F → N → K

Visited: A, B, C, D, G, I, L, M



# BFS Traversal

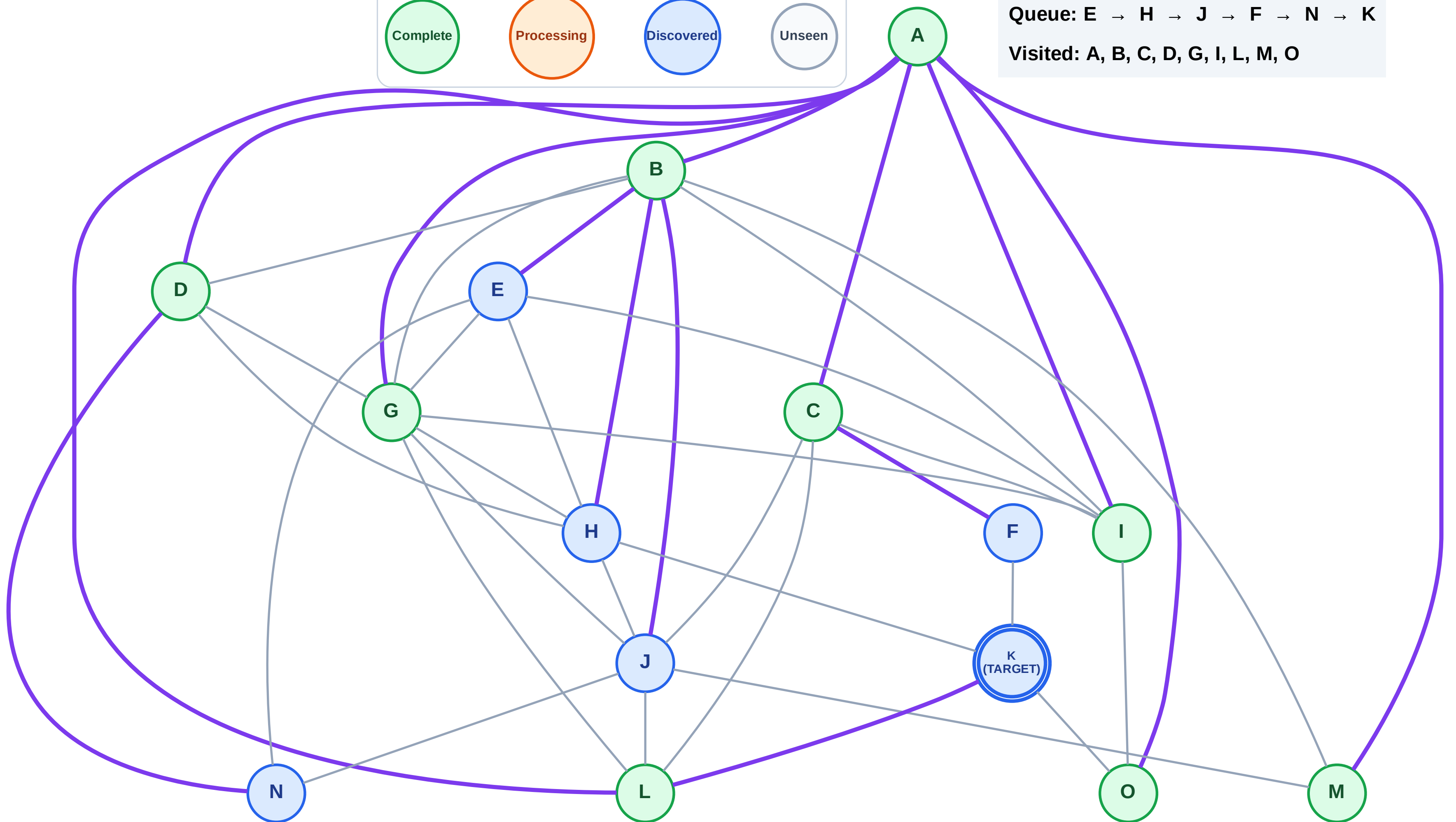
Step 19: No new neighbors from O

## Legend



Queue: E → H → J → F → N → K

Visited: A, B, C, D, G, I, L, M, O



# BFS Traversal

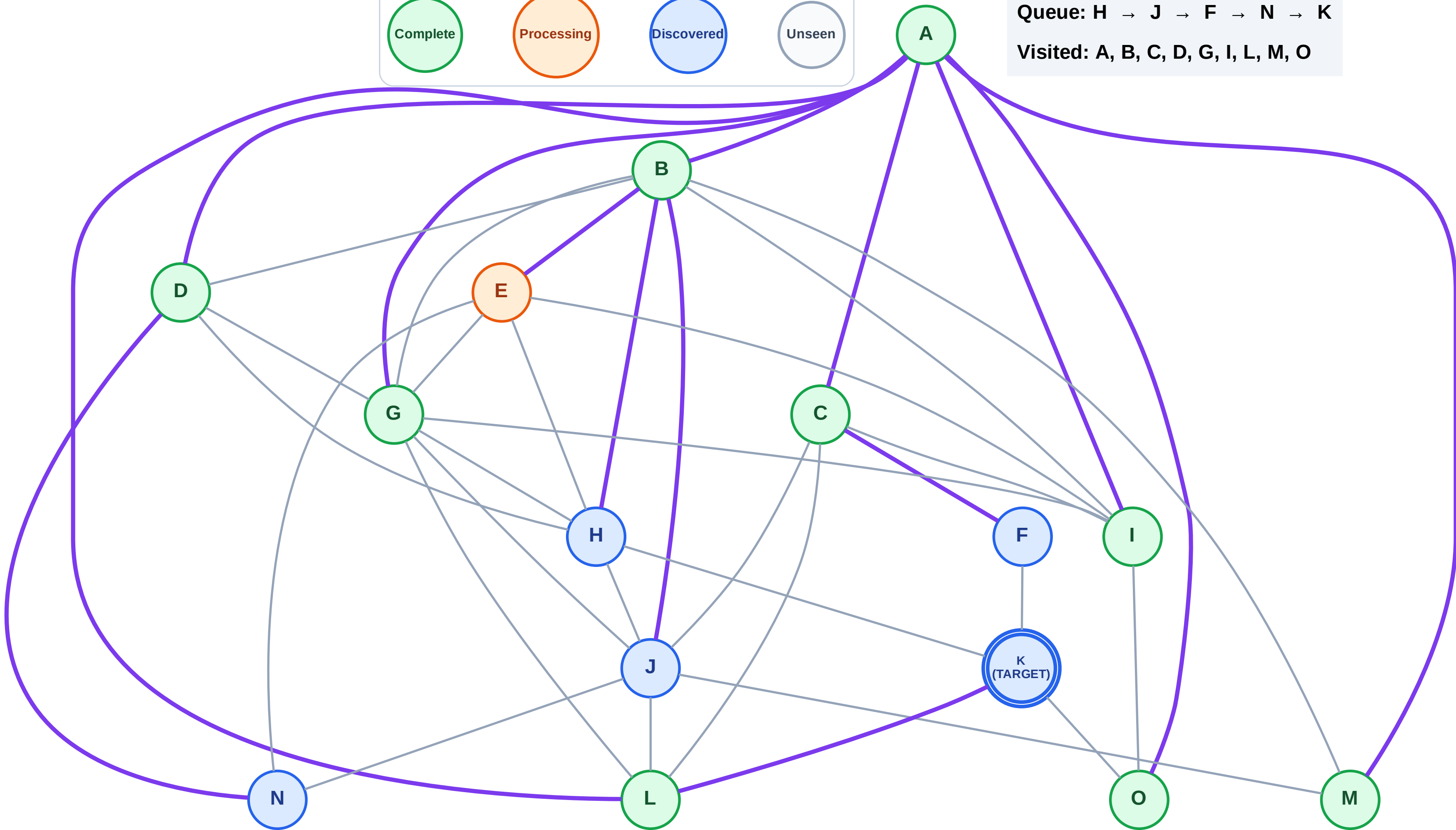
Step 20: Process node E

## Legend



Queue: H → J → F → N → K

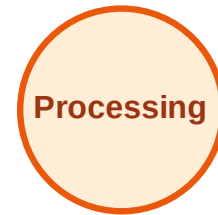
Visited: A, B, C, D, G, I, L, M, O



# BFS Traversal

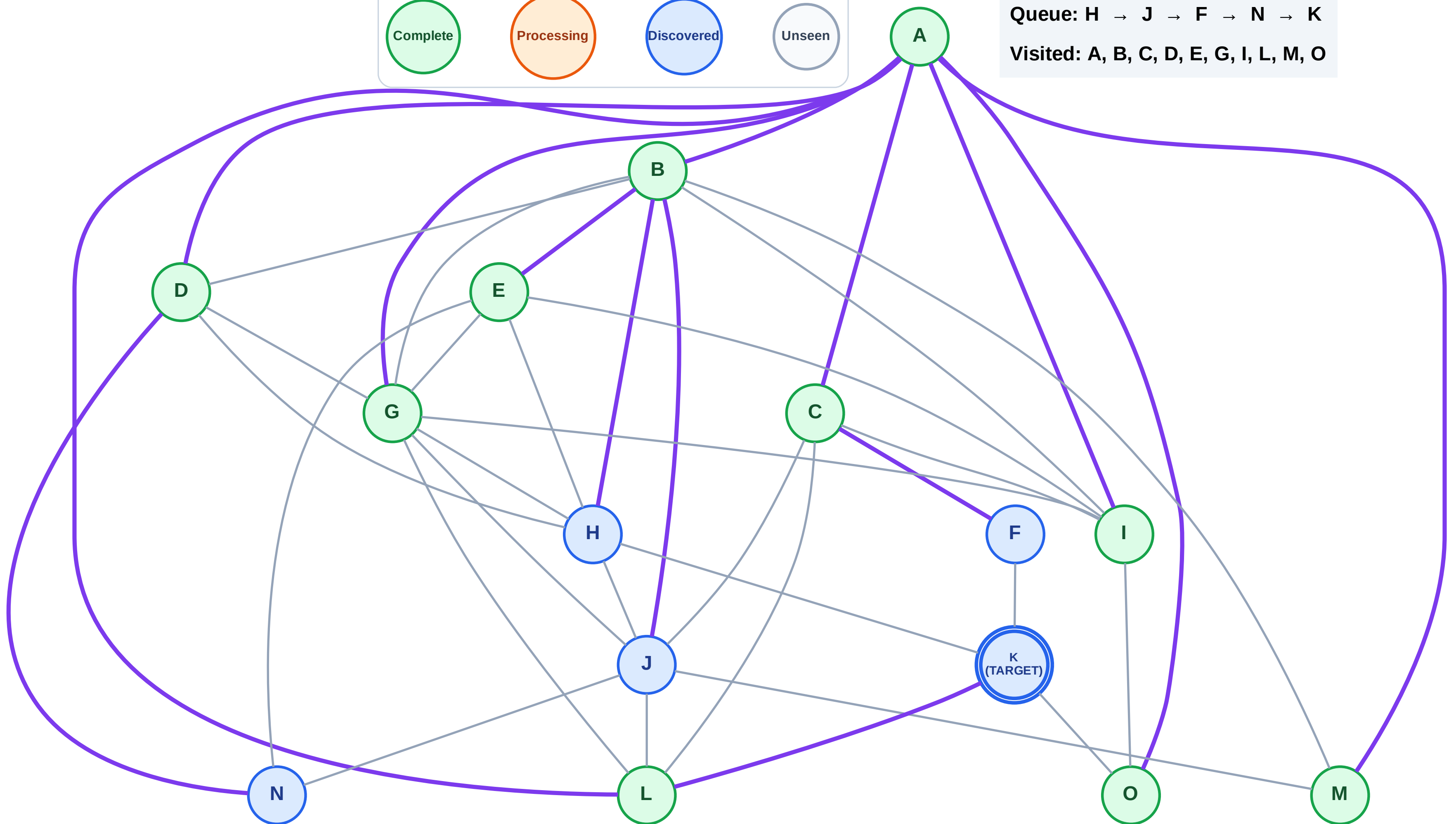
Step 21: No new neighbors from E

## Legend



Queue: H → J → F → N → K

Visited: A, B, C, D, E, G, I, L, M, O



# BFS Traversal

Step 22: Process node H

Legend

Complete

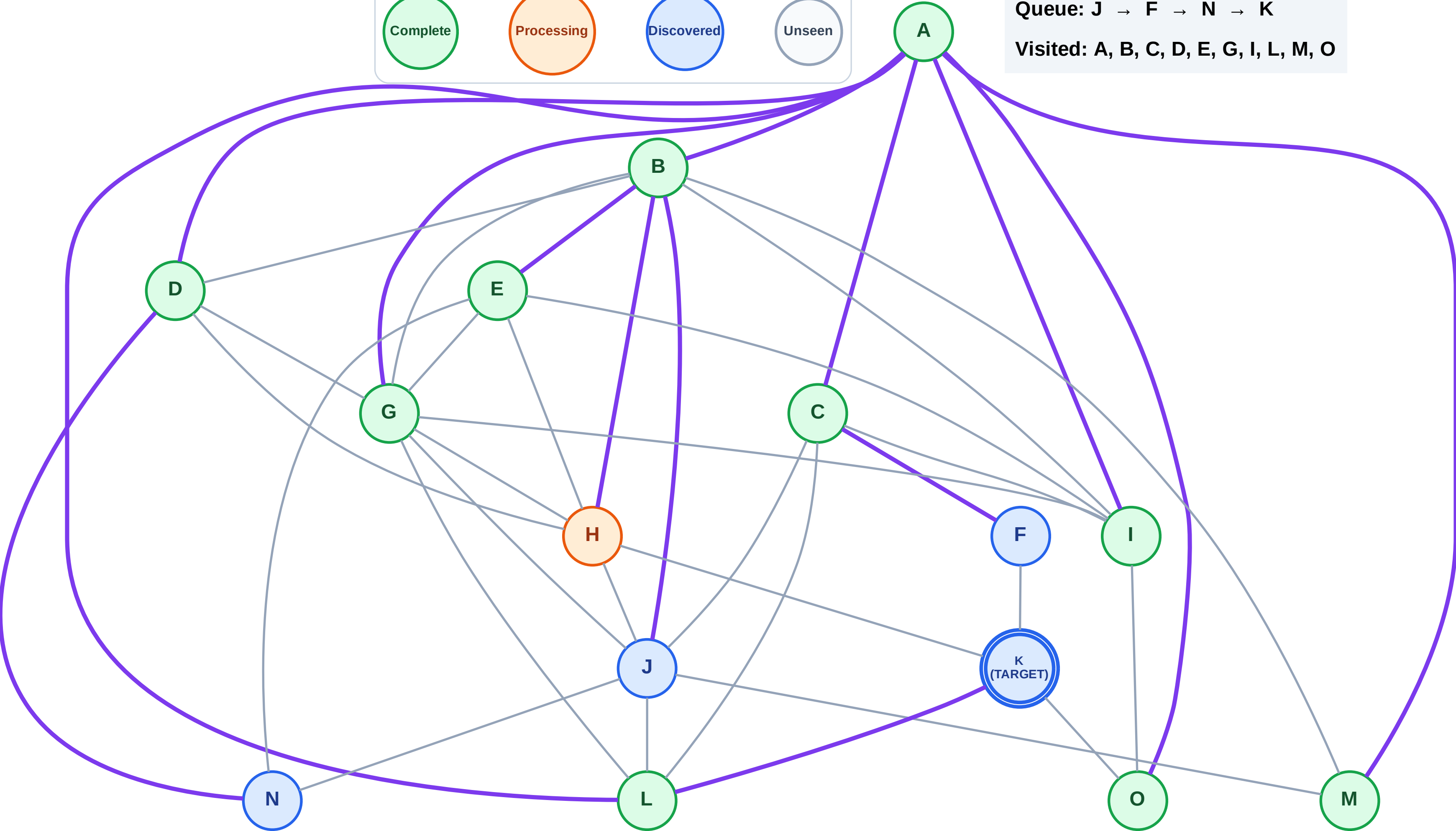
Processing

Discovered

Unseen

Queue: J → F → N → K

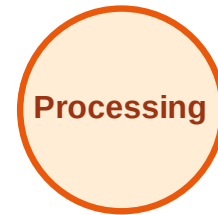
Visited: A, B, C, D, E, G, I, L, M, O



# BFS Traversal

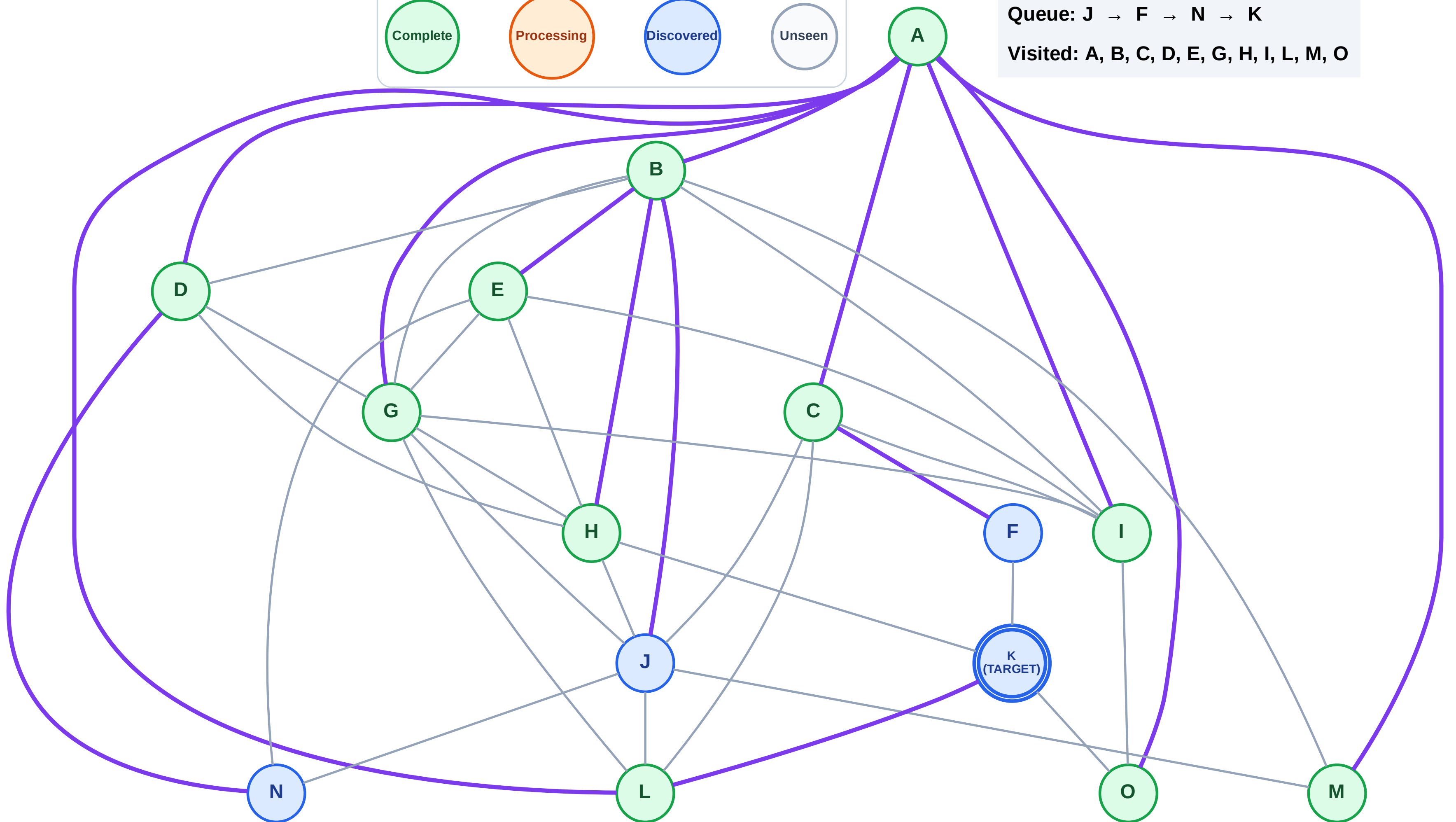
Step 23: No new neighbors from H

## Legend



Queue: J → F → N → K

Visited: A, B, C, D, E, G, H, I, L, M, O



# BFS Traversal

Step 24: Process node J

Legend

Complete

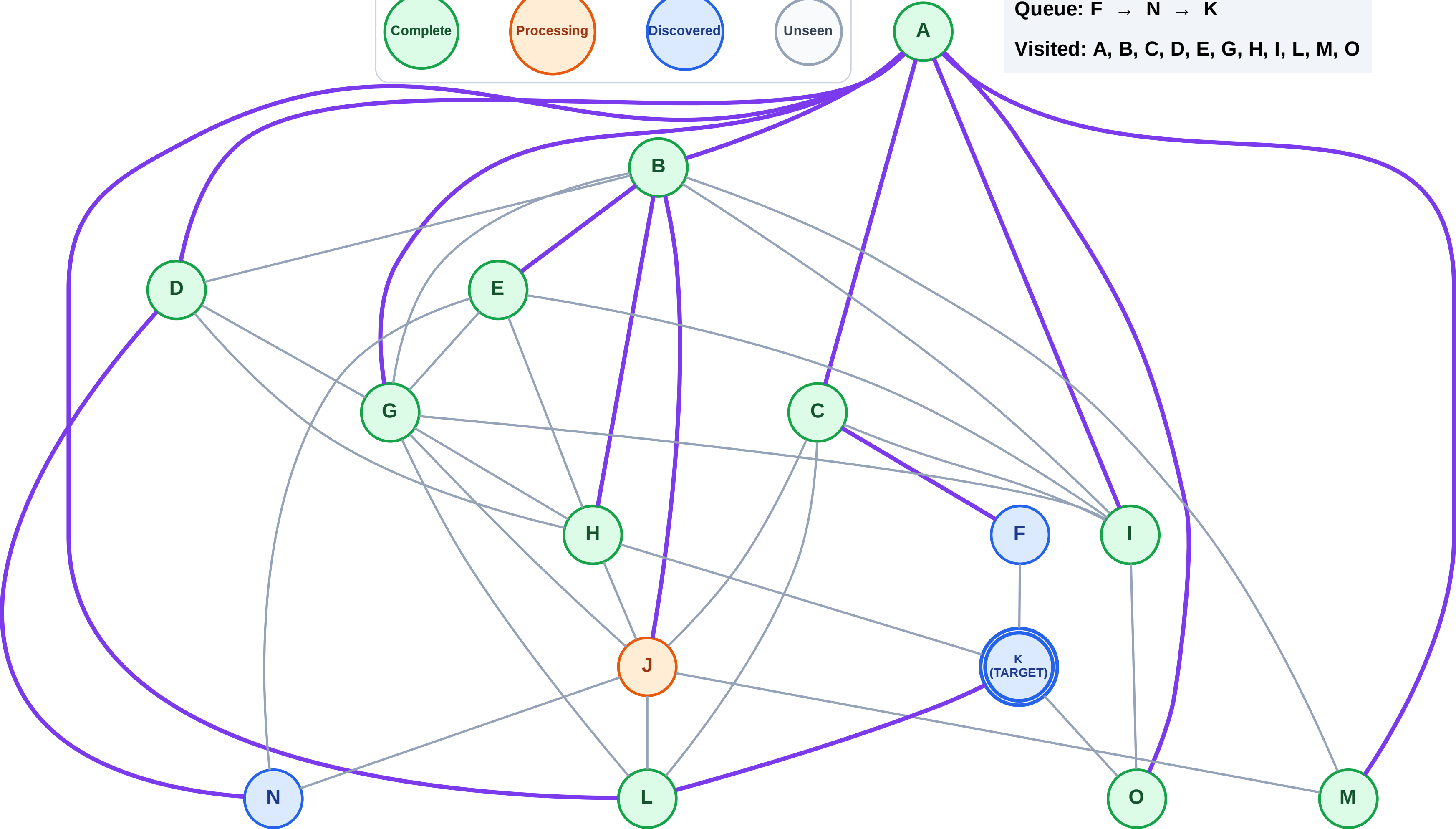
Processing

Discovered

Unseen

Queue: F → N → K

Visited: A, B, C, D, E, G, H, I, L, M, O



# BFS Traversal

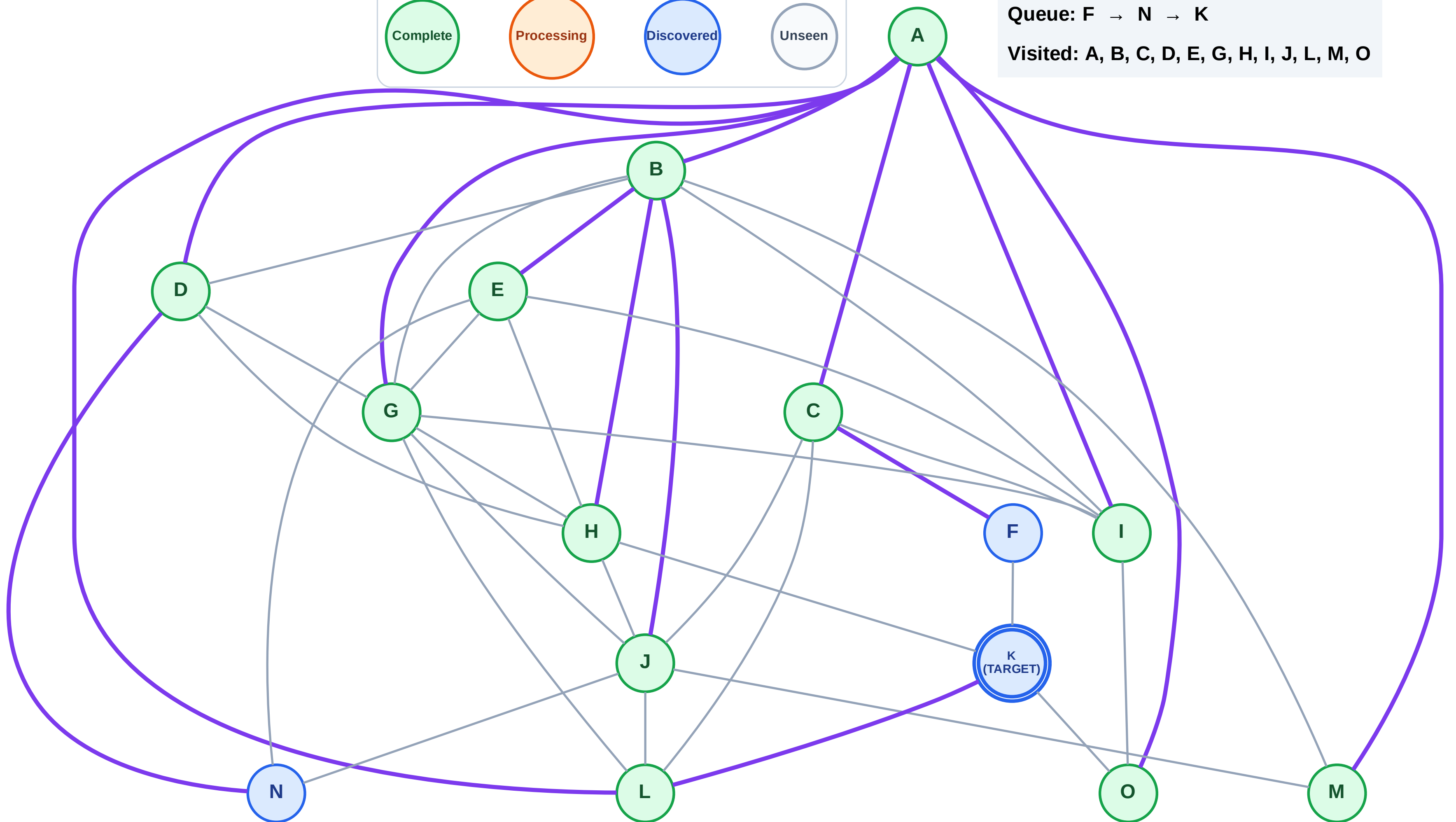
Step 25: No new neighbors from J

## Legend



Queue: F → N → K

Visited: A, B, C, D, E, G, H, I, J, L, M, O



# BFS Traversal

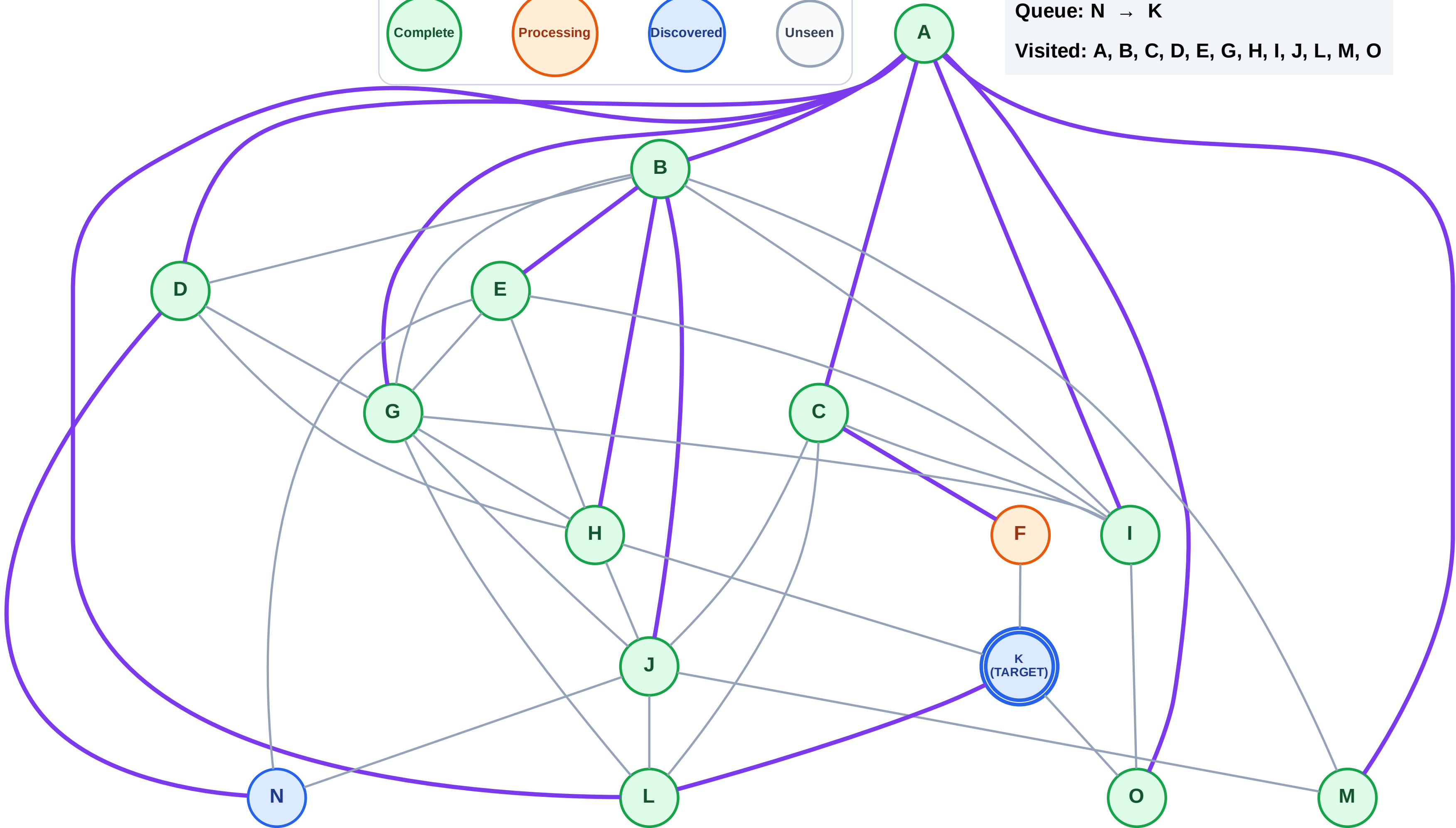
Step 26: Process node F

## Legend



Queue: N → K

Visited: A, B, C, D, E, G, H, I, J, L, M, O



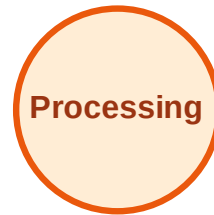
# BFS Traversal

Step 27: No new neighbors from F

## Legend



Complete



Processing



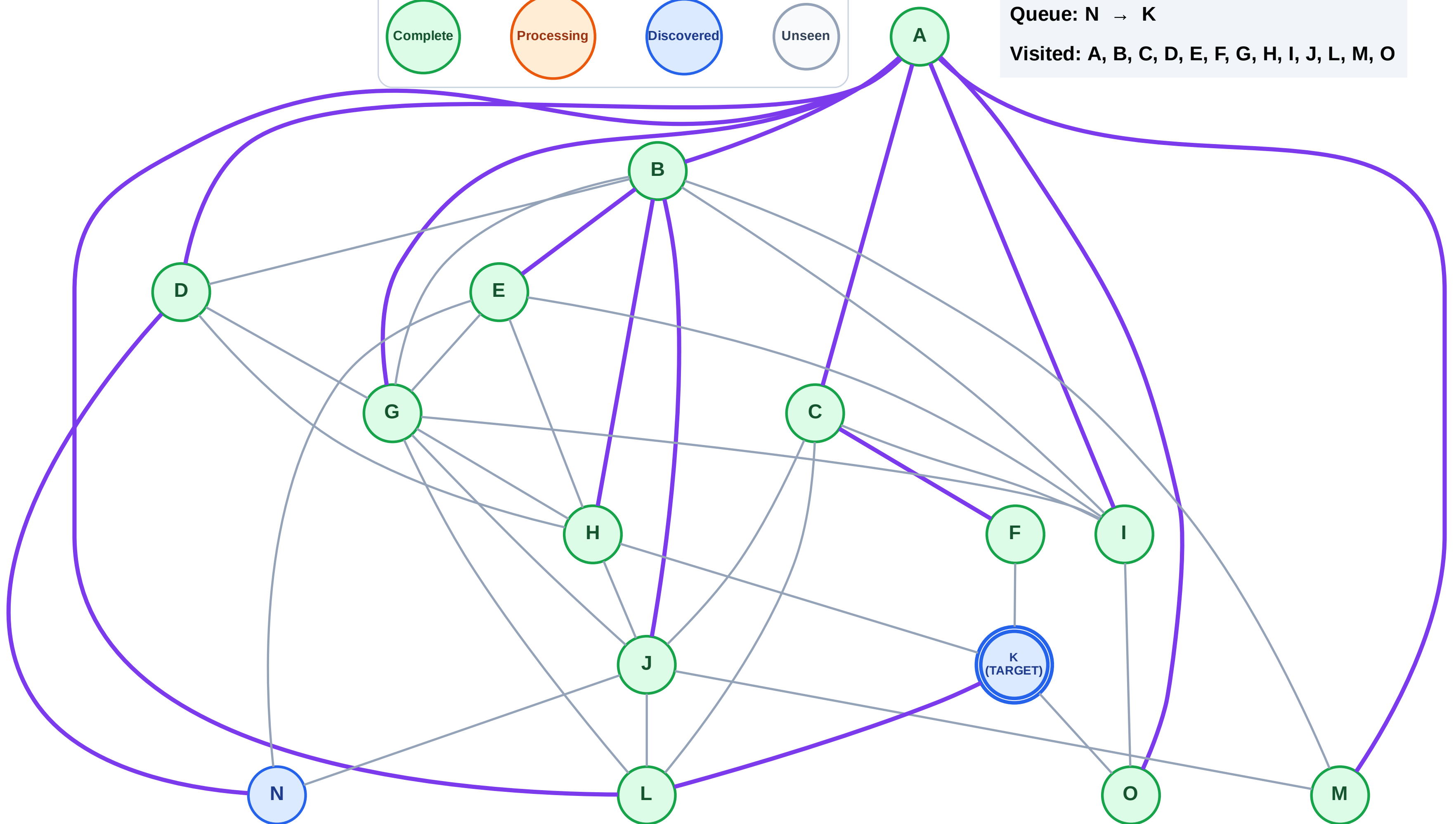
Discovered



Unseen

Queue: N → K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



# BFS Traversal

Step 28: Process node N

Legend

Complete

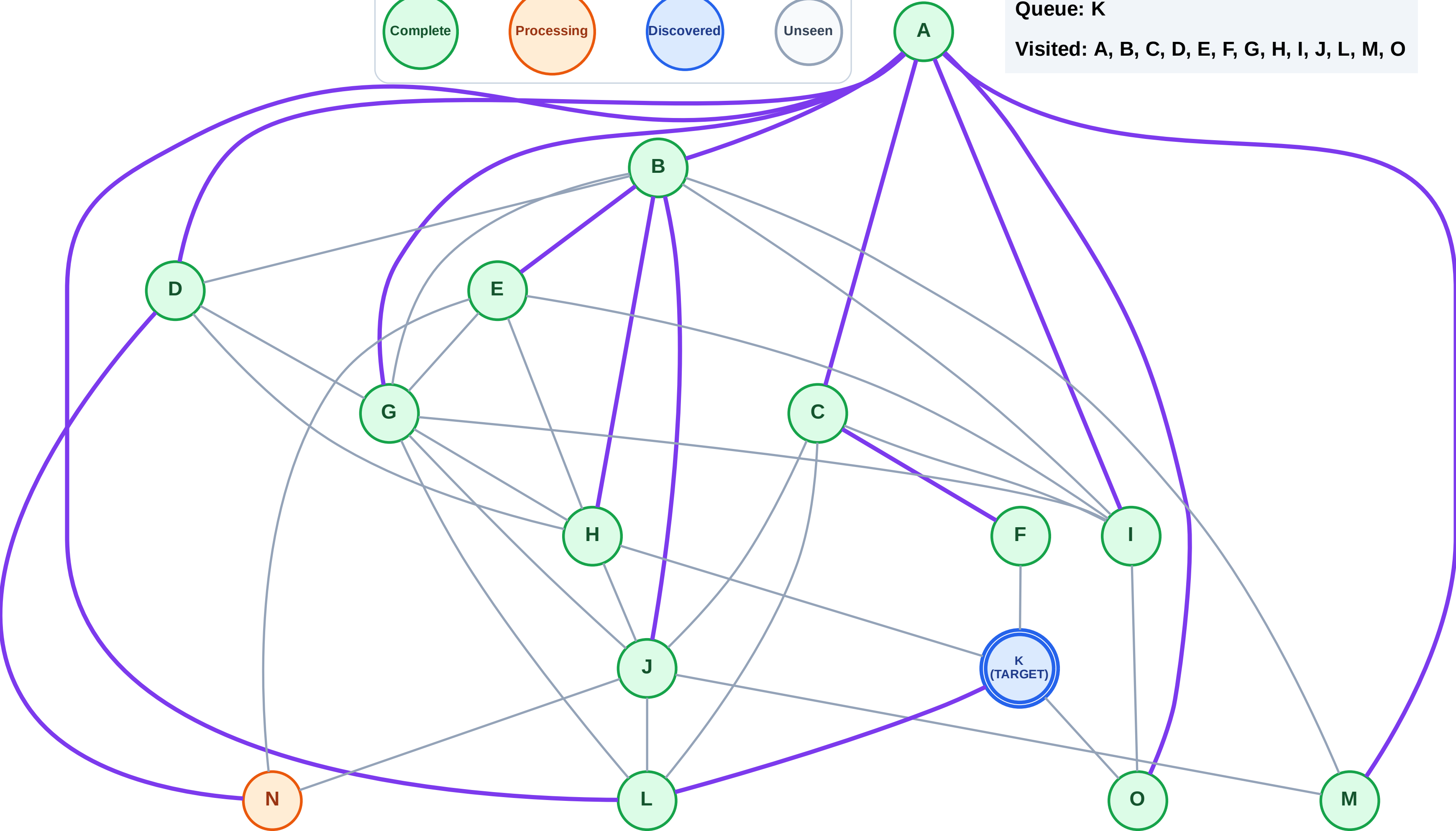
Processing

Discovered

Unseen

Queue: K

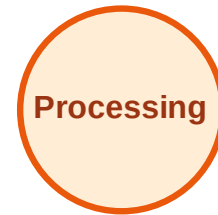
Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



# BFS Traversal

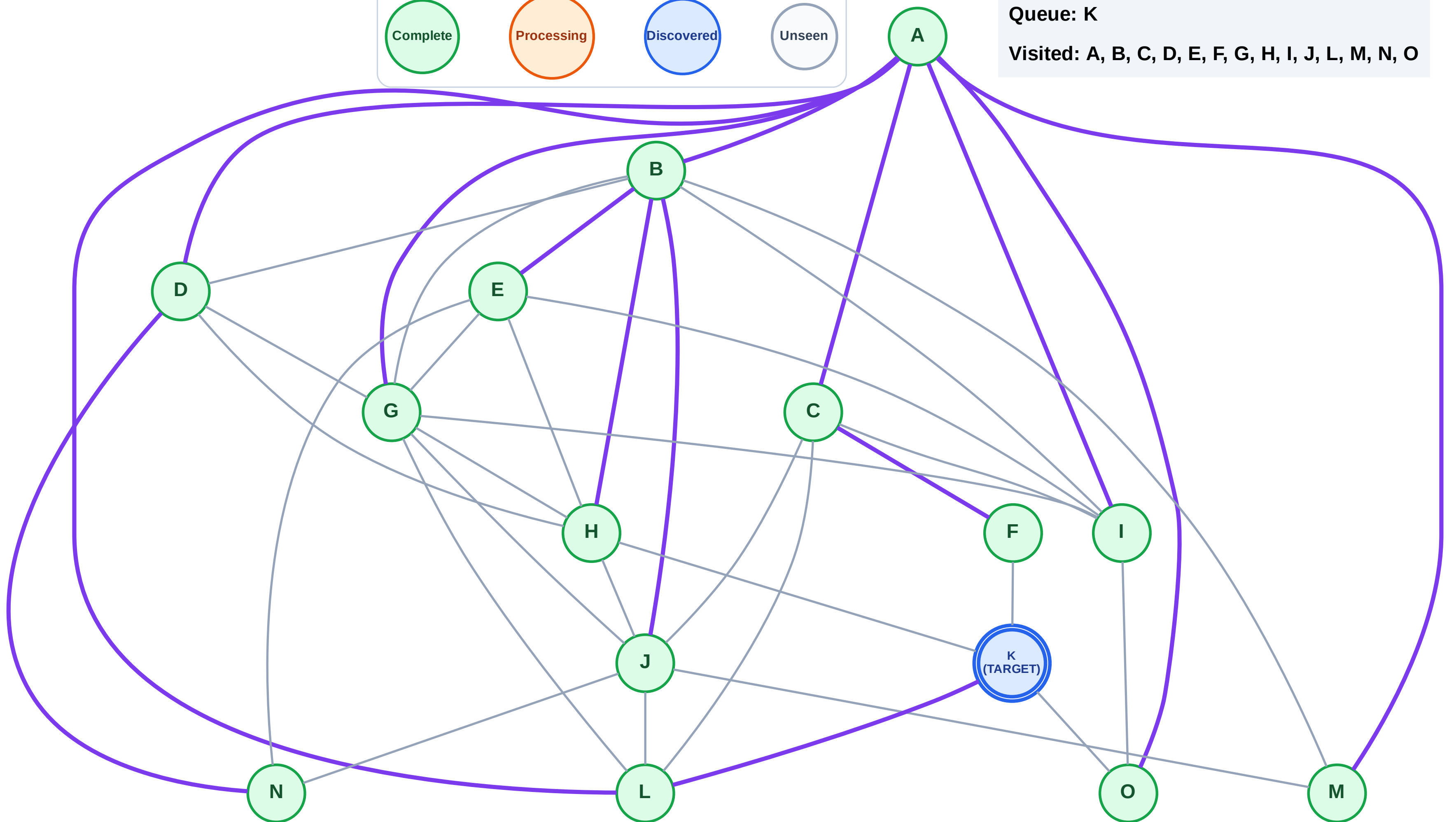
Step 29: No new neighbors from N

## Legend



Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



# BFS Traversal

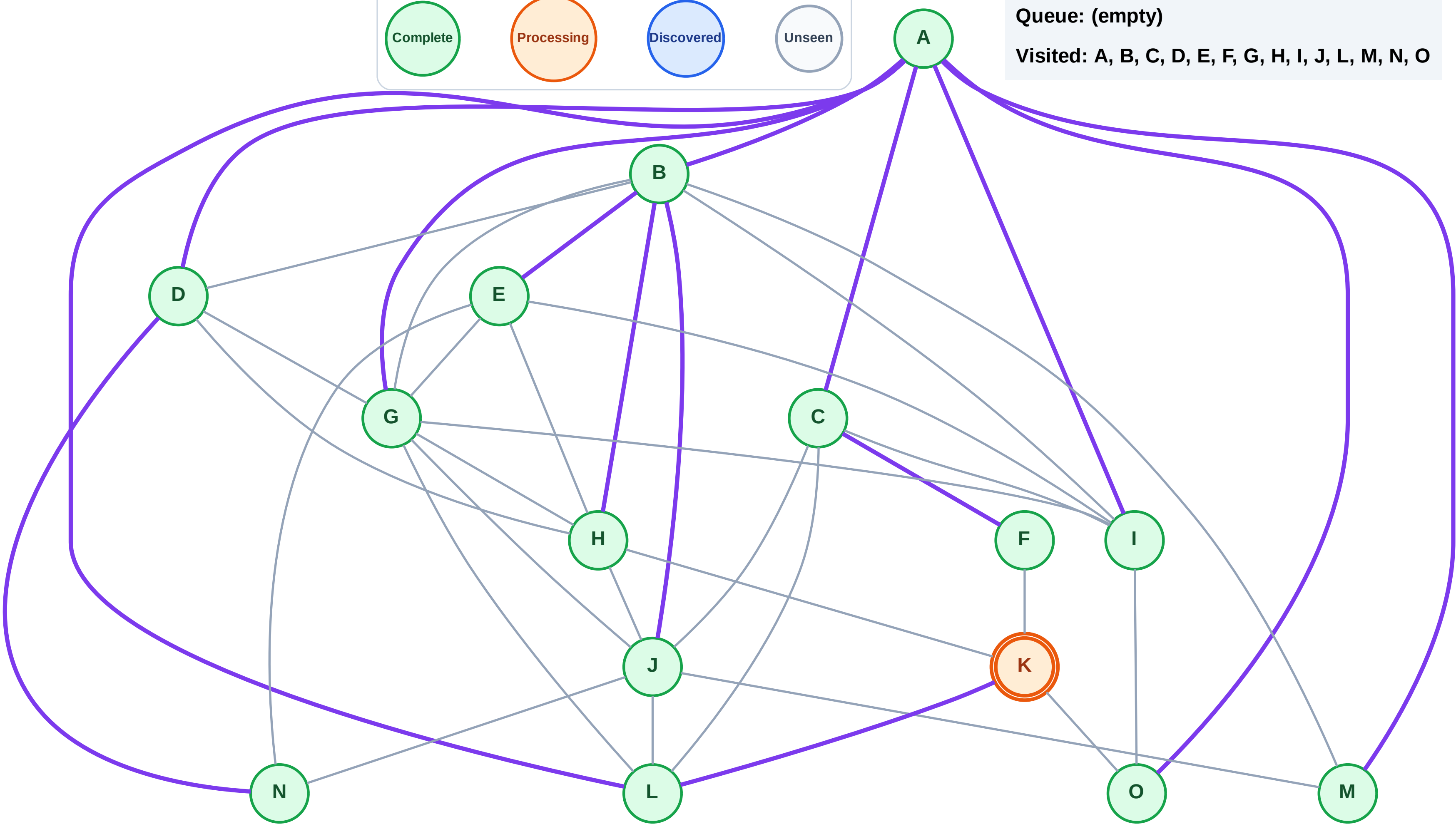
Step 30: Process node K

## Legend



Queue: (empty)

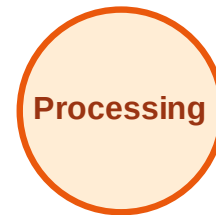
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



# BFS Traversal

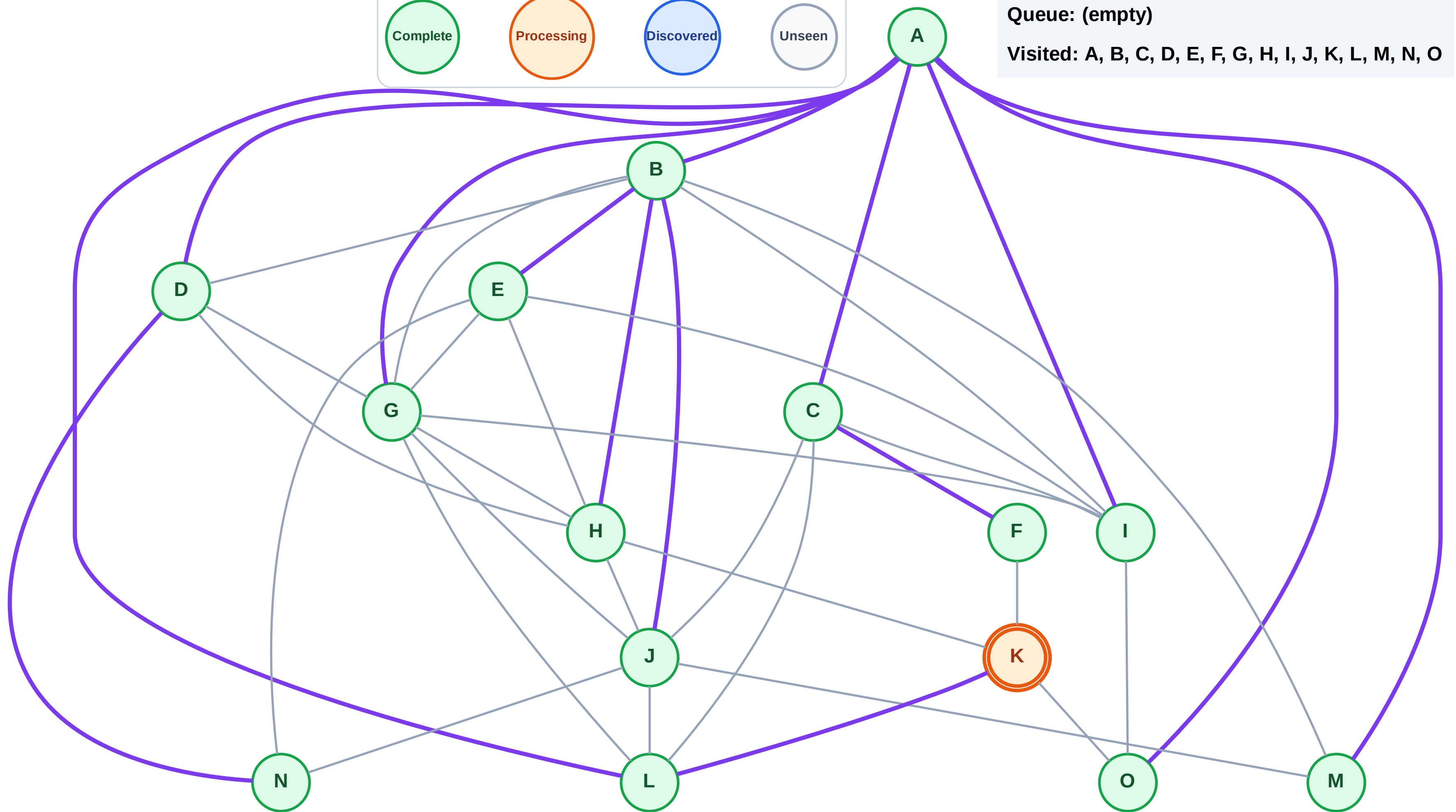
Step 31: TARGET NODE K REACHED! Path (2 hops): A → L → K

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



# DFS Traversal

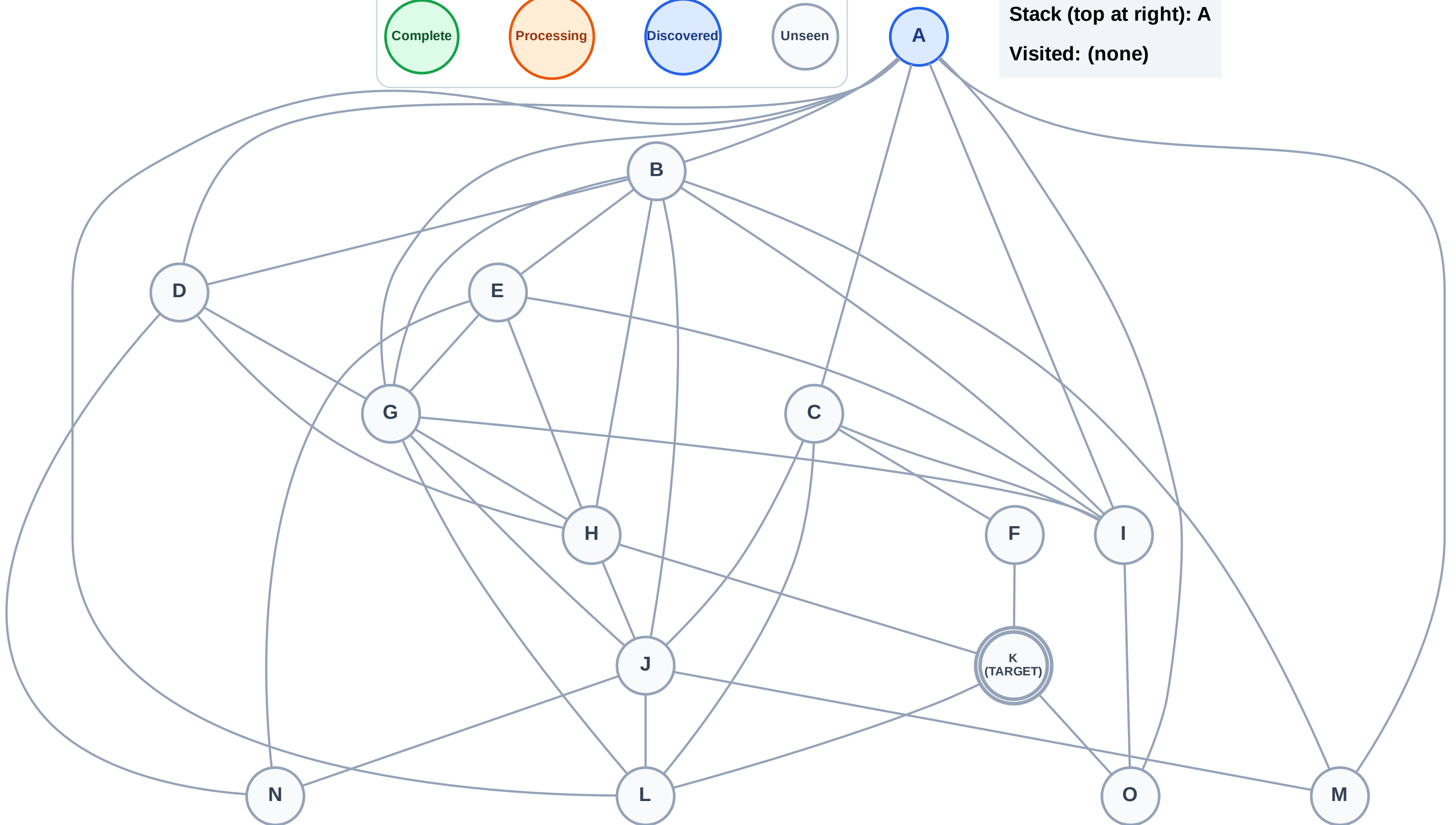
Step 1: Start at A (Seeking target node K)

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

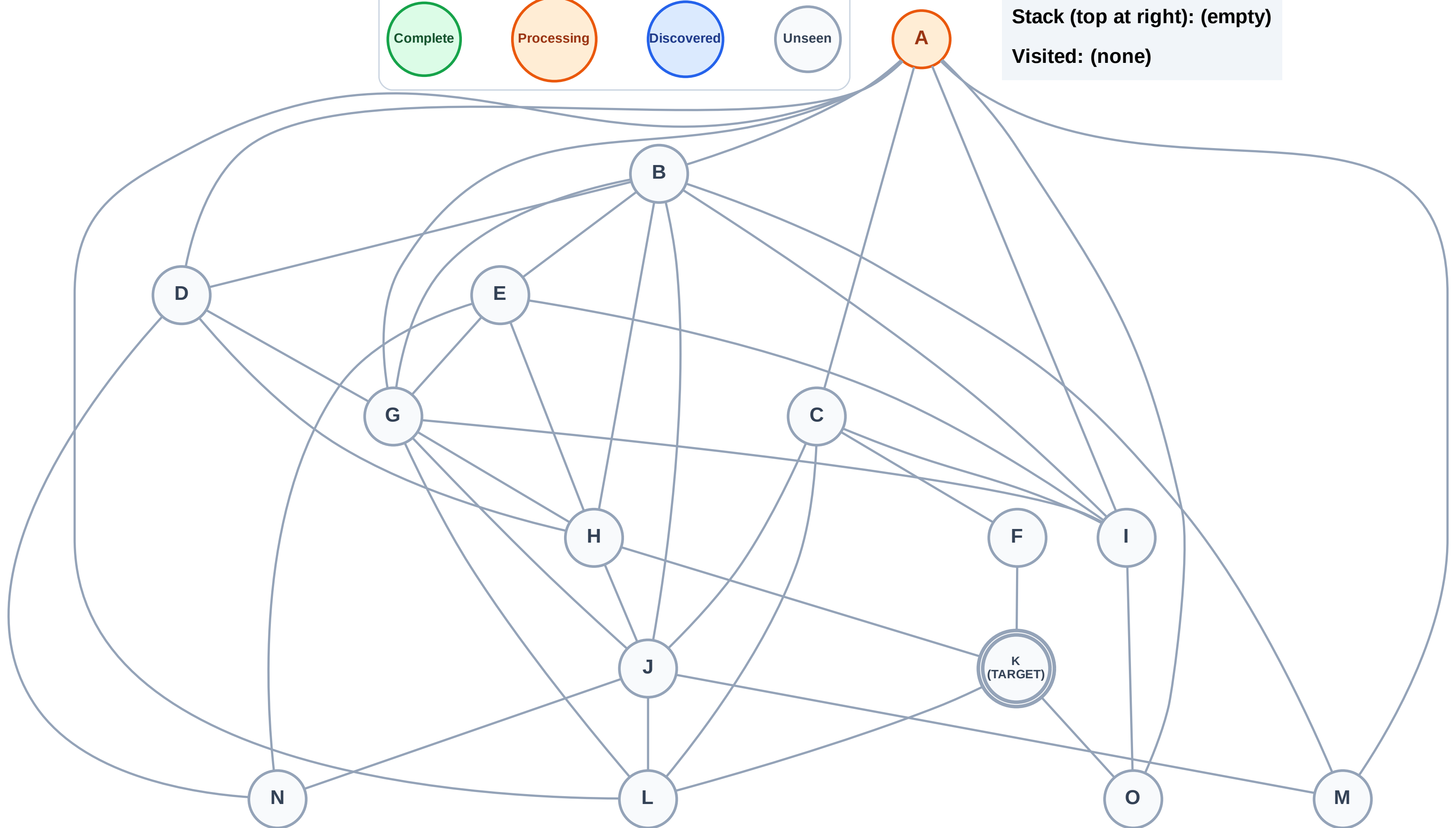
Step 2: Process node A

## Legend



Stack (top at right): (empty)

Visited: (none)



# DFS Traversal

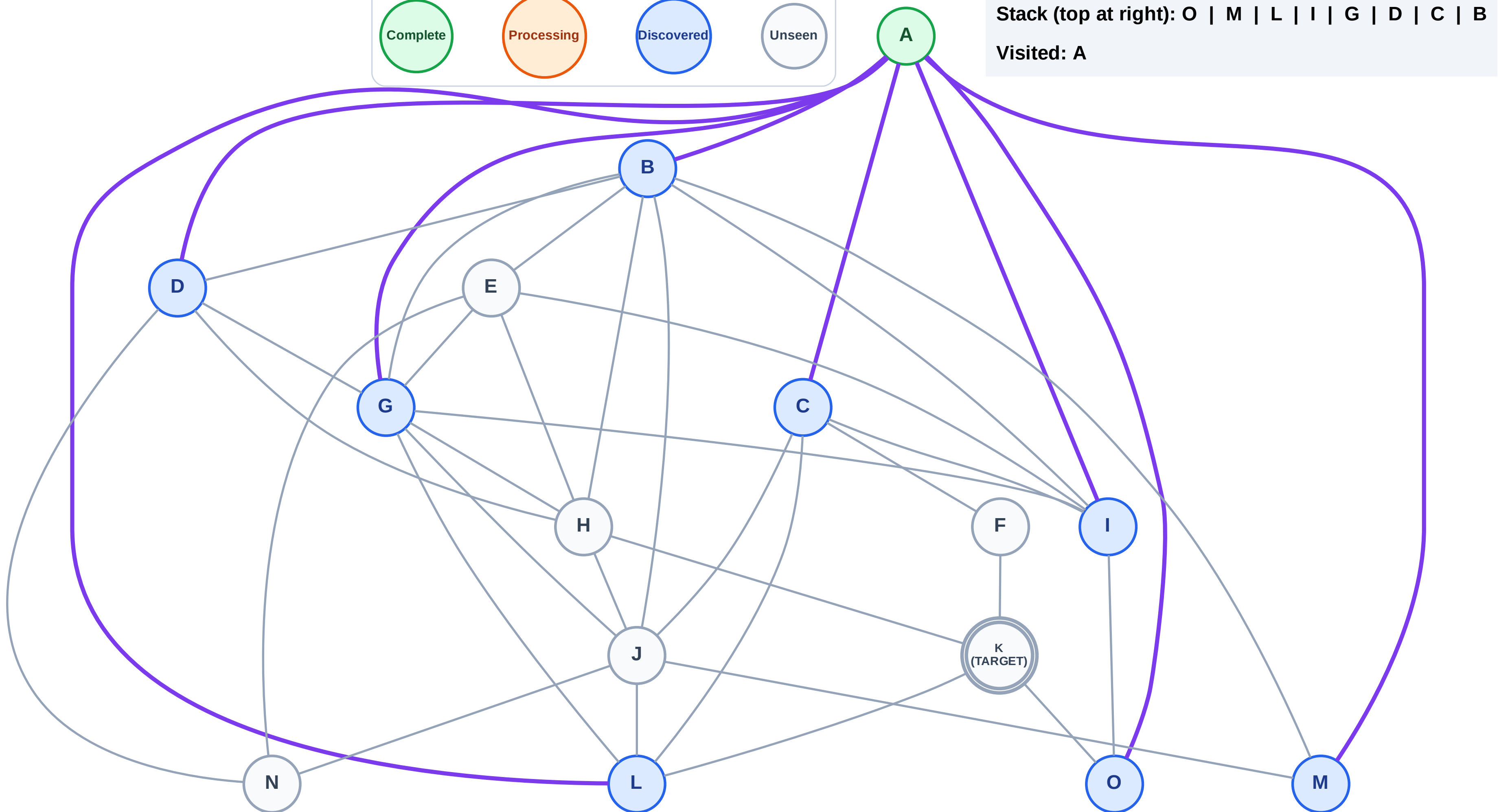
Step 3: Discover O, M, L, I, G, D, C, B from A

## Legend



Stack (top at right): O | M | L | I | G | D | C | B

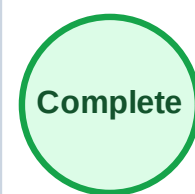
Visited: A



# DFS Traversal

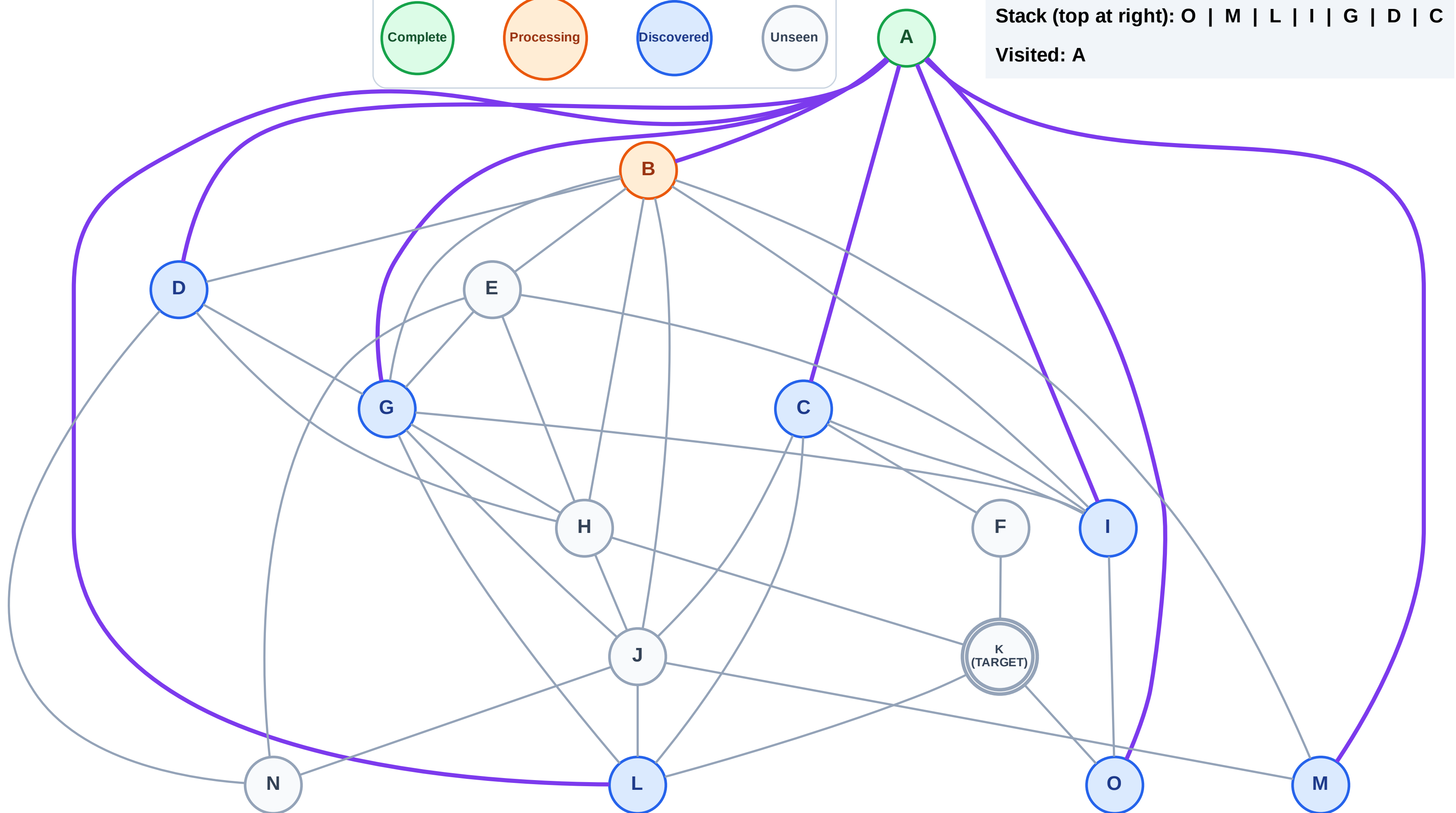
Step 4: Process node B

## Legend



Stack (top at right): O | M | L | I | G | D | C

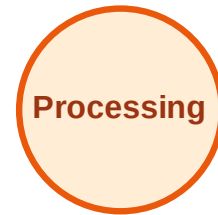
Visited: A



# DFS Traversal

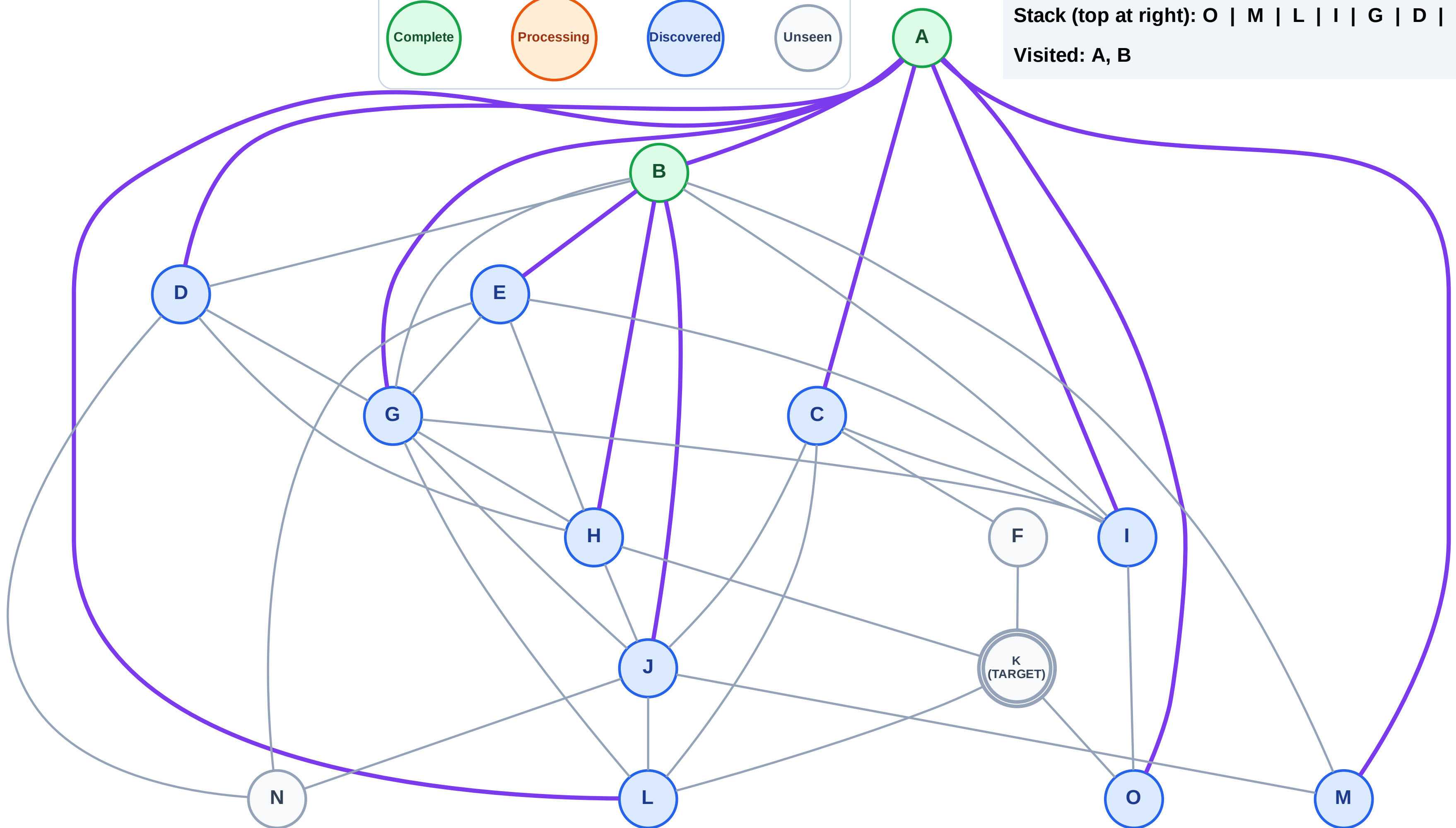
Step 5: Discover J, H, E from B

## Legend



Stack (top at right): O | M | L | I | G | D | C | J | H | E

Visited: A, B



# DFS Traversal

Step 6: Process node E

Legend

Complete

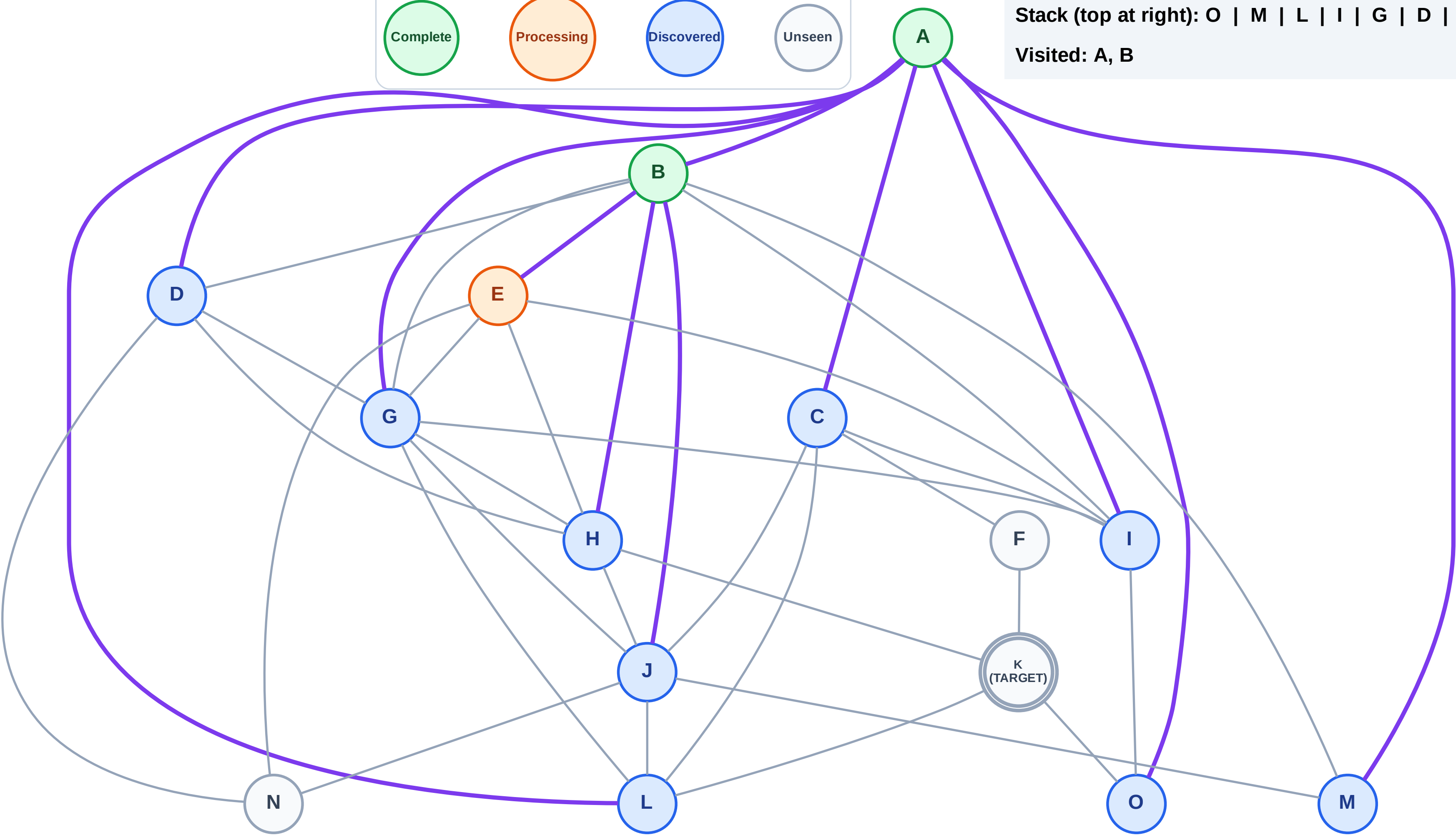
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | G | D | C | J | H

Visited: A, B



# DFS Traversal

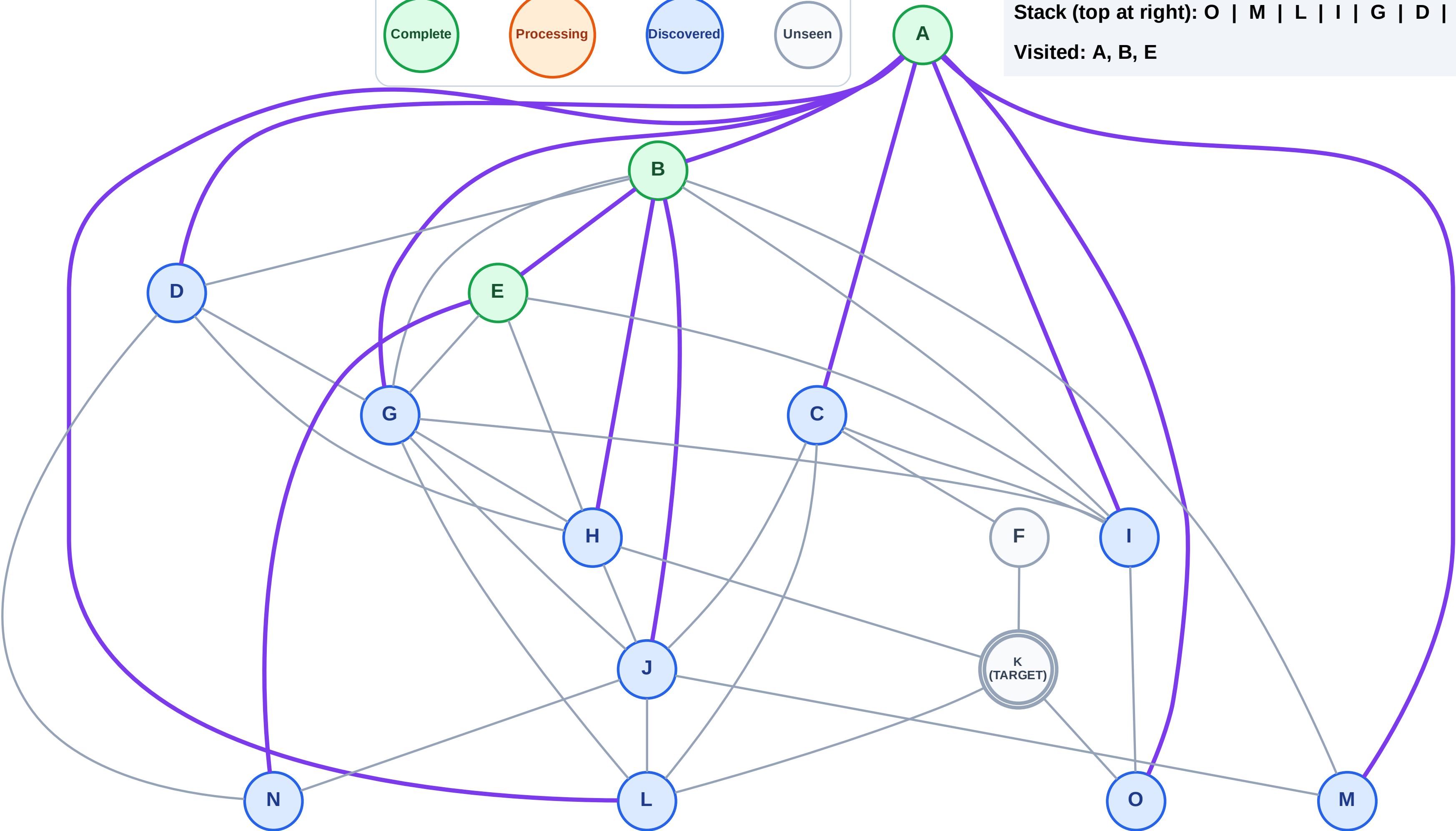
Step 7: Discover N from E

## Legend



Stack (top at right): O | M | L | I | G | D | C | J | H | N

Visited: A, B, E



# DFS Traversal

Step 8: Process node N

Legend

Complete

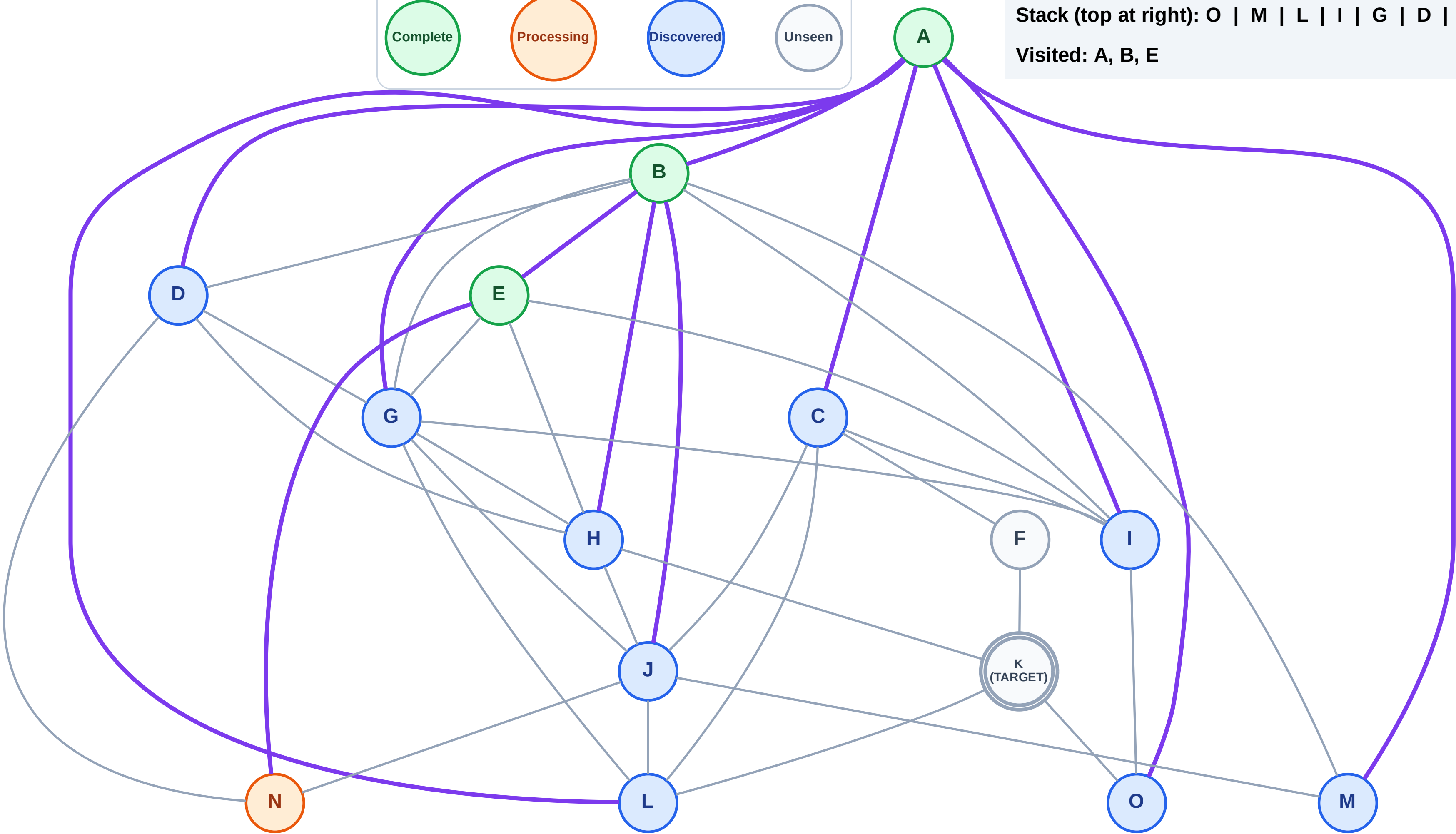
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | G | D | C | J | H

Visited: A, B, E



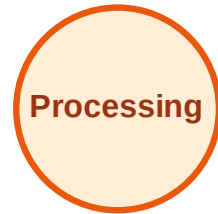
# DFS Traversal

Step 9: No new neighbors from N

## Legend



Complete



Processing



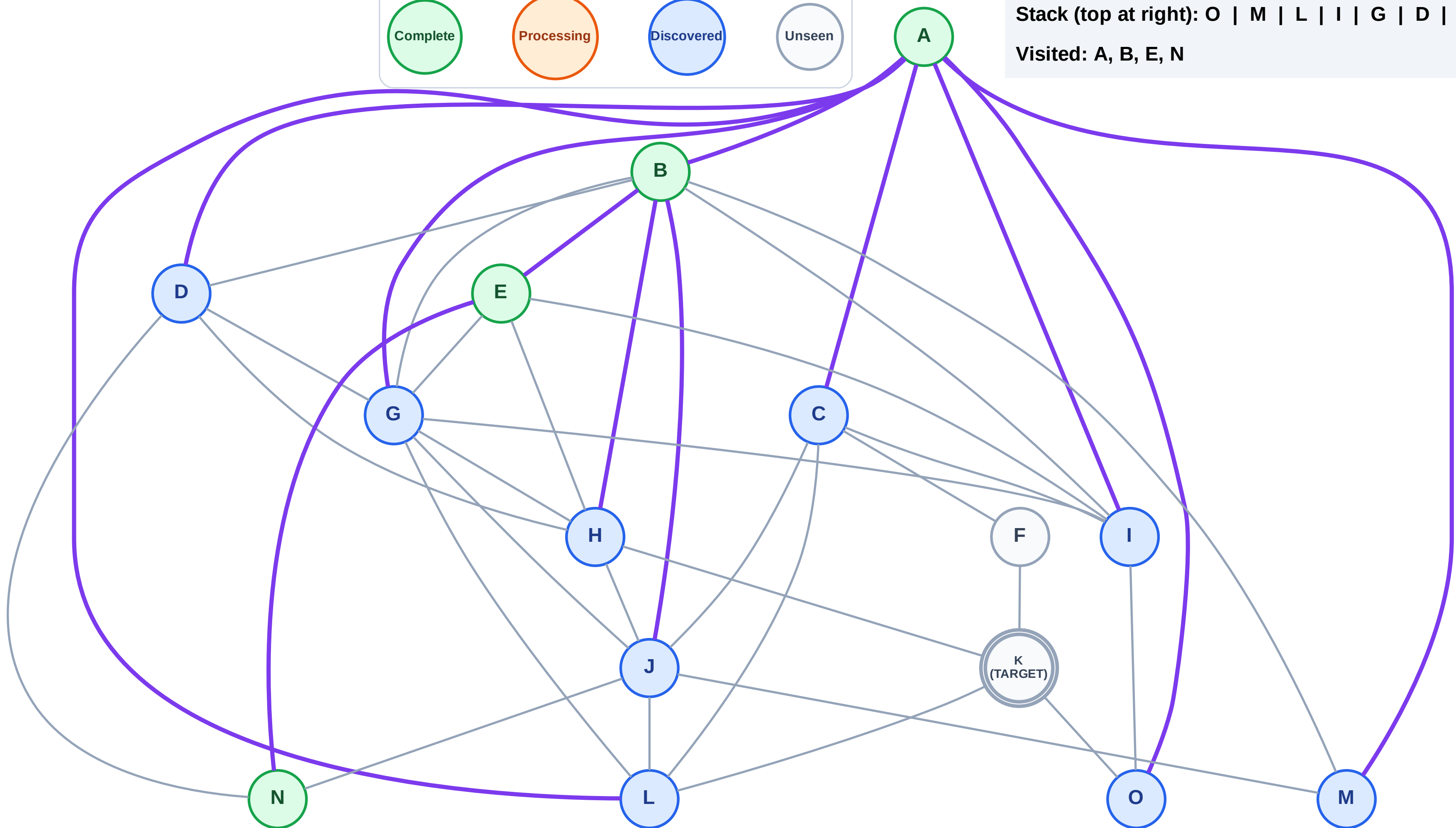
Discovered



Unseen

Stack (top at right): O | M | L | I | G | D | C | J | H

Visited: A, B, E, N



# DFS Traversal

Step 10: Process node H

## Legend



Complete



Processing



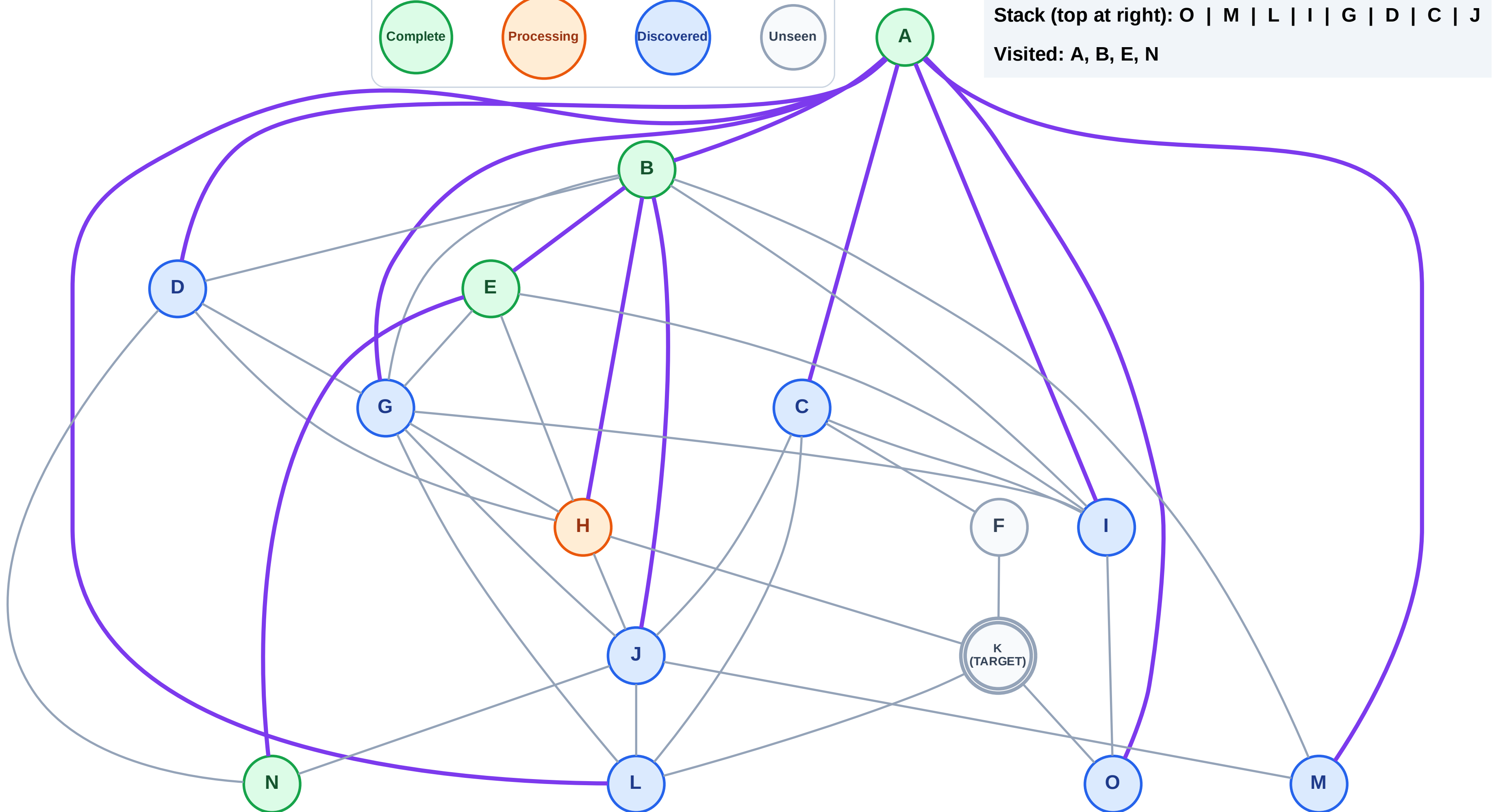
Discovered



Unseen

Stack (top at right): O | M | L | I | G | D | C | J

Visited: A, B, E, N



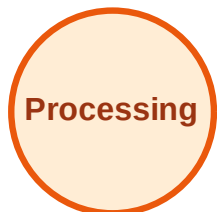
# DFS Traversal

Step 11: Discover K from H

## Legend



Complete



Processing



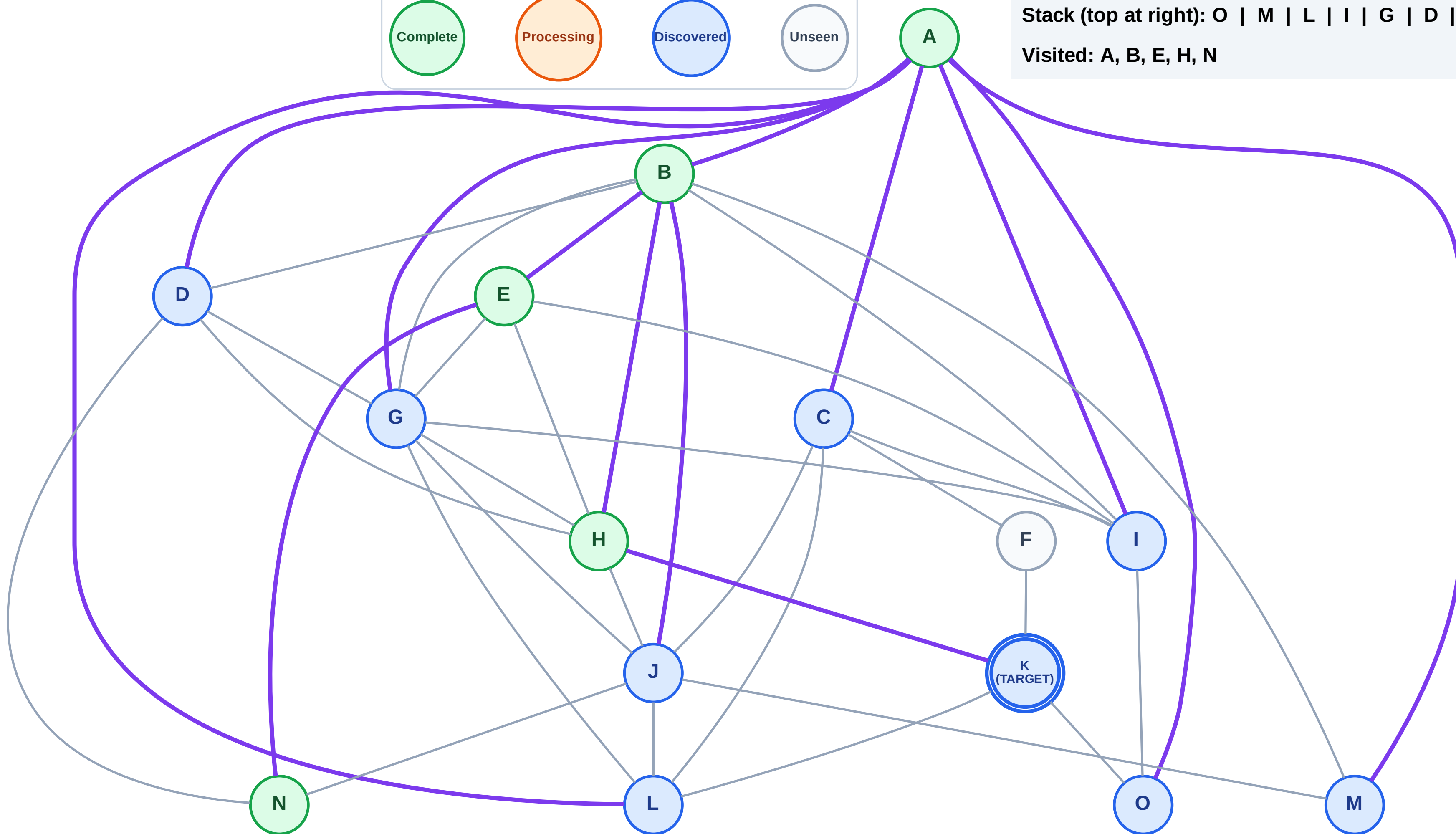
Discovered



Unseen

Stack (top at right): O | M | L | I | G | D | C | J | K

Visited: A, B, E, H, N



# DFS Traversal

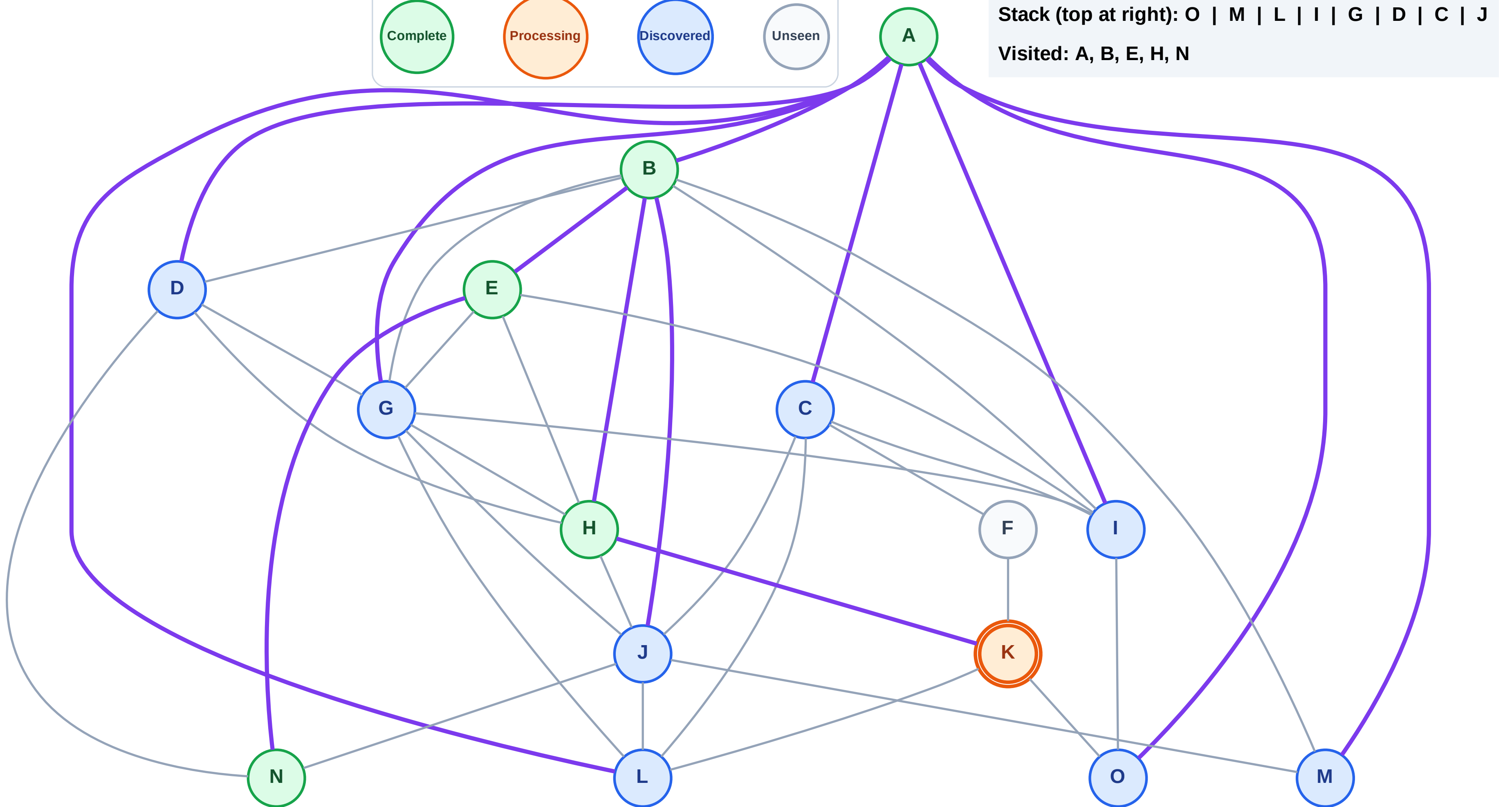
Step 12: Process node K

## Legend



Stack (top at right): O | M | L | I | G | D | C | J

Visited: A, B, E, H, N



# DFS Traversal

Step 13: TARGET NODE K REACHED! Path (3 hops): A → B → H → K

## Legend



Stack (top at right): O | M | L | I | G | D | C | J

Visited: A, B, E, H, K, N

